NEO COMPUTER FAMILY · SPECIFICATION AND DEVELOPMENT PLAN · WORKING DRAFT 1 · ENGLISH (UK)

NeoOS 1.0

What goes into the first version of the operating system, how it is built from the kernel up to the icons, and how many people and months it takes to get there, apps and games included.

~41 mo.
from start to 1.0, realistic case (March 2027 to August 2030)

~148
people at peak, 10 teams

5,350
person-months in total, design and management included

33 + 9
apps and games in 1.0, plus 6 FPGA cores

5 November 2026 · concept · every figure is an engineering estimate by analogy with similar projects, not a measurement or a quote. Builds on "Neo Inside" (the technical book) and the register of adopted decisions. Neo names are given in their English UI form; the Serbian original is in the glossary at the end.

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PART A · SPECIFICATION

1. What NeoOS 1.0 is

NeoOS 1.0 is the first version shipped to customers on all three models (Neo 500, 2000 and 4000), running from the Neo ROM, with every in-box app marked P0 and P1 and the launch games. Version 1.0 is a promise that the system can be the only computer in the house: mail, web, office, photos, music, learning, games and retro, with no account and no telemetry.

Scope: what is in and what waits

In 1.0

- Neo card with Autoconfig and portability to RISC-V (part 7b)
- Crunching: crunched programs with no unpacker inside, data crunched on the fly (zstd, LZ4, xz), the Disk tile with its "Crunch" button, in Lite too (part 7d)
- Fortress with native drivers for all Neo hardware, Exec-N, Dispatcher, Chronos, Renewal, Lock and Safe
- Full graphics stack: Vulkan 1.4, OpenGL 4.6 and ES 3.2, Glide, Direct3D through Bridge
- Neo Silver look with classic Mac OS, BeOS/Haiku and MorphOS elements (part 8)
- Device and user profiles, Serbian (Latin and Cyrillic) and English (en-GB and en-US), translation kit (part 1b)
- Learning before games: Neo School, Classroom, Playroom, Neo Blocks, Lab, Sky, Recall; News from sources the user picks (part 11a)
- Working with tasks and documents after Lisa, Xerox Star and Canon Cat (part 8b)
- Shelves, Helpers, everyday language and Dijana as resident and guide (parts 8 and 8c)
- Beauty without burden, system sound, files in Easy mode and Co-op (part 8d)
- Retro and Future Shock styles, ideas from the Unix world (part 8e)
- A computer for people: Paper, e-ink, Focus, breaks (part 8g)
- Phone alongside Neo, the Neo MCP bridge for Claude and other AI (off until switched on), touch mode (part 8h)
- Tiles (small programs: clock, calendar, meter, notes, weather, battery, calculator), the Bar, transparent if you like, the Neo menu and desktop layouts (part 8i)

Waits for 1.1 and later

- Android room (Waydroid with microG) and ReactOS box (experimental)
- Neo Workshop, the Pro pack (part 10)
- Pocket profile for handhelds (with Neo OS Lite)
- Neo 100 and Neo OS Lite (1.1), Neo Revive for old x86 and ARM computers (1.5), part 7c
- Flow across several Neo computers in the house
- Neo Studio, Neo Cards, Cut and Broadcast (P2)
- Export from Neo Studio to Android and iOS
- Neo OS Lite for phones (experimental, 2.0, part 7c)
- Our own metroidvania, Ras, if the licence falls through

- Neo DOS: the native shell in the AmigaDOS spirit, on an equal footing with the Desk (LoadDesk, EndDesk, Assign), a Linux shell for experienced users and the Translator between them (part 8j)
- Cabinet Pro: the Duo layout in the spirit of Directory Opus 4 and Norton Commander, programmable button banks, filetype actions, triggers and user menus after Magellan (part 8k)
- Even easier: input by task, Guest USB, Call your people, Progress, shared parts, Forever format, First week (part 8f)
- Bridge (x86 programs and Windows games), Linux box, Satellite on 2000 and 4000
- Arcade with six FPGA cores and emulators
- Local voice in Serbian (both scripts) and English; Ask Neo with Library
- 33 apps (P0 and P1), 9 games, Treasury, SDK and Neo BASIC

Target measures for 1.0

Every measure has a lab test. If a measure is not met, 1.0 does not ship.

Measure	Target	How it is measured
Cold boot to desktop	under 6 s (Neo 4000), under 8 s (Neo 500)	camera on the screen, from key press
Resume from a Chronos snapshot	under 1.5 s	same
Mouse-to-screen latency	under 8 ms at 120 Hz, 99th percentile	photo sensor and Courier timestamps
Sound	1 ms buffer with no dropouts for 72 h under full load	Orpheus missed-buffer counter
In-box app launch	under 300 ms to first paint	Exec-N trace
RAM at idle	under 1.5 GB with Cabinet open, not counting the RAD: fast layer and the system held in RAM	Gate statistics
System image in ROM	under 12 GB compressed (fits the 16 GB pSLC on Neo 500)	build
Driver or domain restart after a crash	under 2 s, no open documents lost	deliberate crash in test
Update and rollback	switch to the new image in under 20 s, roll back with one button	A/B test over 500 cycles
Graphics	Vulkan 1.4 CTS pass, OpenGL 4.6 and ES 3.2 CTS pass through Zink	Khronos tests
Hardware support	every built-in device works through a native driver, with every capability in the technical book	test matrix per model
Ease of learning	a new user sends mail, writes a letter and finds a photo within 10 minutes, without a manual	test with 30 people new to Neo
Localisation	100% Serbian (Latin and Cyrillic) and English (UK and US)	string check

> Hardware is the key assumption: the plan expects the first Neo package silicon (chiplets with VSA-6000) around month 24. Until then the OS is developed on a development platform: a fast off-the-shelf ARM computer (Snapdragon X class), an FPGA card over PCIe and VSA-6000 in pre-silicon FPGA emulation.

1b. One OS for everyone, on every device

NeoOS is one system with a human face: the same for a child, an adult and a professional, and the same from a handheld to a workstation. Only the device profile and the user profile change, never the system.

Device profiles

Profile	Device	What changes
Pocket	handheld and phone (experimental)	Neo OS Lite, touch mode, Retro and Calm
Light	Neo 100, old computers (Revive)	Neo OS Lite, emulators instead of FPGA
Handheld Pro	powerful handheld on a Neo card	full NeoOS, touch mode, Arcade with FPGA, battery saving
Home	Neo 500 and Neo 2000	full NeoOS
Workstation	Neo 4000	full NeoOS, Workshop, several monitors, Full throttle

User profiles

- **Kids' room**: Easy mode, learning first, a parent sets time and content, no purchases and no outside messages.
- **Easy**: adults who want the job done; everyday names and Dijana as guide.
- **Large**: for older and partially sighted people: big text and targets, fewer things on screen, “Call your people” in one tap.
- **Pro**: technical names, Cabinet Pro, Shell, Lux and every option.
- **Shared computer**: for libraries, schools and community centres: each guest gets a clean system, and “Good night” wipes it all.

Languages: Serbian and English first

- **Serbian** is equal and stressed: Latin and Cyrillic as two complete languages, automatic transliteration, Serbian cases and genders in system messages (“3 fajla”, “5 fajlova”, “Vojinov računar”), spell checking, voice and dictation.
- **English** in two complete variants, British and American (en-GB and en-US): spelling, dates, units, A4 or Letter paper, dictionaries and voice. Even the names follow: Bin in en-GB, Trash in en-US.
- The language changes at once, without a restart; each member of the household can have their own.

Translation kit (for any country and language)

- **Project Fluent** (Mozilla, Apache licence) as the message format: it handles the cases, genders and plurals of any language, so translations into Slavic, Finno-Ugric or Semitic languages are not boxed in by English grammar.
- **Weblate** (GPL) as a tool a government or community hosts itself: translators work in the browser, see screenshots next to each string, and changes reach Treasury.
- Pseudo-localisation to check that text fits buttons, CLDR data for dates and numbers, keyboard layouts, right-to-left scripts.
- A “New language in 90 days” guide: what to translate first (system, Quartet, School) and how to add voice, dictionary and Interpreter.

```
The same privacy promise holds for every country and every edition: no
hidden tracking and no exceptions for authorities.
```

2. Layered architecture

NeoOS has seven layers. The bottom three are the guards: they decide who may do what. The top four are the workers: everything the user sees is there. The rule is simple: a layer above can never bypass the layer below.

Apps and games	Cabinet · Horizon · Quartet · Arcade · Pinball Cosmos · programs from Treasury
Neo Kit and Lux	controls, layout, Easy/Pro, themes, Datatypes, ports and scripts
System services (Rust)	Exec-N · Dispatcher · Chronos · Renewal · Lock · Keeper · Chameleon · Find
Native drivers (Rust), Linux domains	drivers for all Neo hardware · Linux only for third-party peripherals, file systems and the program ABI
Fortress	seL4-class microkernel (EL2), Gate: hands out capabilities and memory
Neo ROM and Safe	boot loader in NOR, A/B image in pSLC, signatures and keys
Hardware	ARM · FPGA/CGRA · VSA-6000 · Ray · Newton · Panorama · Orpheus · Copernicus · Neuron · Courier · Chronos · Sat

Language and speed

The system is written in Rust, and hand-written ARM assembly (SVE2) is used only for the hottest 1 to 2% of code: memory copies, cryptography, codecs and blitter routines, measured before and after. An OS written entirely in assembly would not be noticeably faster, because the speed comes from the design (no copying, everything in RAM, accelerators, priority for input and sound), and it would be several times slower to build, less secure and tied to one generation of ARM.

Four design rules

1. **Responsiveness before throughput.** Input, sound and window drawing have a guaranteed share of CPU time in Fortress. Nothing in the background may stall the mouse.
2. **One system, one API, one look.** Neo Kit is the only windowing toolkit; foreign programs (Linux, Windows) get the same frames through the compositor and Link.
3. **Every app is scriptable.** A Lux port in every program, like ARexx on the Amiga.
4. **Small and understandable.** Every system app has a memory and size budget, after SymbOS. Going over budget is a build error.

3. Kernel: Fortress

Fortress is an seL4-class microkernel: small (about 10,000 lines), formally verified code that does only three things. It shares out memory, schedules processor time and passes messages between the protected parts of the system. Above it sit **native drivers for all Neo hardware**, written in Rust as separate processes, and the system services. Linux remains in only a few confined places, with no access to built-in hardware. When anything above the kernel crashes, Fortress restarts it and the rest of the system carries on.

How it is put together

Part	Where it runs	What it does
Microkernel	ARM EL2 (hypervisor level)	seL4 with MCS scheduling, and above it the Genode component framework (section 12b): every thread has a budget and a period, so sound and input always get their time. Virtualisation through ARM hardware for the Linux domains.
Gate	root service, Rust	The only party that hands out capabilities: which domain may use which memory, which device, which channel. It manages the IOMMU (SMMUv3) tables, so every device sees only its own memory.
Native services	seL4 processes, Rust	Exec-N, Dispatcher, Chronos, Renewal, Lock, Safe broker, Keeper. Small, Linux-free, each in its own address space.
Native drivers	seL4 processes, Rust	Every device on the Neo board has a native NeoOS driver, both our own chips and bought-in components. The hardware is fixed, so drivers are written once and tested on exactly three models. The list is below.
Linux domains	virtual machines on Fortress	Only three, and none touches built-in hardware: third-party devices the user plugs in, file system code, and the Linux ABI for programs.

Native drivers for Neo hardware

The rule for 1.0: everything on the board or in the Neo package works through a native NeoOS driver, with full support for every feature, not just basic operation. No built-in device depends on a Linux driver.

Group	Devices	Note
Graphics and display	VSA-6000 (all chiplets, T-Buffer, Glide mode), Ray (RT, blitter, compositor), Panorama (6 outputs, RAMDAC, 15 kHz, drag-down screens, HDR, VRR)	Kernel part native in Rust; the Vulkan part (Mesa "vsa") runs in the program's process and sends work to the driver through command buffers
Our own chips	Copernicus, Newton, Neuron, Chronos controller, Courier (DMA, decompression, video), Safe, FPGA/CGRA (Chameleon)	Written alongside the chip, from the first model in simulation
Sound	Orpheus DSP and analogue section, DAC, headphone amplifier, XLR, S/PDIF, HDMI eARC	A 1 ms buffer is guaranteed by Fortress
Storage and media	NVMe (M.2 and Neo cartridge with FRAM), Neo ROM (NOR and pSLC), UHD Blu-ray/M-DISC drive	Block devices are handed to the Storage domain
Buses and ports	PCIe 5, AGP-X, CXL (Satellite), USB4 and xHCI, DE-9, HDMI input, Neo remote	USB classes (keyboard, mouse, disk, audio, camera, joystick) are native
Network and wireless	2.5G and 10G Ethernet, Wi-Fi 7, Bluetooth 5.4 LE Audio, 5G M.2 modem	We choose chips with public documentation and firmware that may be shipped; the network stack is native, in Rust, and Lock sits on top of it

Group	Devices	Note
Camera and sensors	Eye S, M and Pro (sensor, ISP, autofocus, ToF), microphone array, privacy switches	ISP tuning is done separately for each sensor
Power	power supply, Neo 500 battery, 48 V input, Sun station (MPPT), chiplet power gating, fans	Solar mode is run by the Dispatcher

Three confined Linux domains

Domain	Contains	Hardware access	When it crashes
Peripherals	third-party devices the user plugs in that have no native driver: unusual USB devices, old printers and scanners	only the USB port the native driver passes to it	restart in under 2 s
Storage	file system code: mirrored Btrfs and the EROFS image	none; it receives block devices from the native NVMe driver	restart with a journal check, open files carry on
Applications	the Linux ABI for programs (WebKit, GStreamer, the LibreOffice core), each program in its own sandbox, the Linux box and Bridge	none; graphics, sound, camera and input go over paravirtual channels to the native drivers (virtio-gpu native context, virtio-snd, virtio-input)	Chronos restores the state from a few seconds earlier

Why programs still use the Linux ABI: WebKit, GStreamer, LibreOffice and thousands of libraries expect POSIX and Linux system calls, and writing our own POSIX personality from scratch would add a year. It is not a driver question: the Applications domain sees no device at all and sends work to the native drivers. A game over Vulkan loses less than 3% per frame on that path, which is part of the month-12 gate.

Exec-N across domain boundaries

Messages between programs and services travel through shared-memory rings that Gate approves, without copying (similar to virtio queues). A port has a name (`Cabinet.1`, `Gramophone`), so a Lux script or a voice command does not need to know which domain the program lives in.

```
; Lux: the same works from a script, by voice and from Ask Neo
ADDRESS Cabinet
  OPEN "Pictures:2026/November"
  SELECT DATE yesterday
ADDRESS Album
  SHOW SELECTED
```

Fallback plan

The gate after month 12 checks whether Fortress meets its targets: IPC overhead under 5% on the Cabinet benchmark, graphics overhead under 3% per frame, and a correct restart of every driver and domain. If it does not, Genode moves to a different kernel (NOVA or base-hw), and the last resort is **pKVM** (protected KVM from Android). Either way, native drivers stay native and only the layer beneath them changes. We lose the formal proof of the kernel but keep everything else, and the plan slips by about 2 months instead of 6 or more.

4. System services

Service	Responsibility	Specification for 1.0
Exec-N	messages and ports	Messages up to 64 KB in a ring, larger data as shared buffers. Priorities: real time (sound, input), interactive, background. Message latency under 5 µs within one domain.
Swarm	small jobs	Neo Kit cuts work into small pieces and Swarm spreads them over cores and chips (Newton, Ray, Neuron, Orpheus); a core that finishes takes work from its neighbour. Short jobs go to SIMD (NEON, SVE2, SME), long streams to a chip, because sending to a chip costs a few microseconds. Swarm measures and learns what runs faster where. One shared memory, with no manual copying of data (a lesson from the Cell processor). Native Neo programs do not create threads by hand; ported Linux programs keep their own threads.
Dispatcher	where a program runs	Four classes: **1 at hand** (a reserved core, at most one frame from touch to picture), **2 heard and seen** (guaranteed time on Orpheus and Panorama), **3 background** (pauses at once when needed), **4 asleep**. The rule: background work never steals a frame. **Full throttle** mode (the Turbo key) gives the big cores, graphics and part of the cache to one program, and everything else moves to the efficiency cores. It chooses ARM, x86 translation, Satellite or an FPGA core according to the program's manifest. It powers down unused chiplets. Solar mode moves heavy jobs to times of surplus power.
Chronos	state and time	Program snapshots, 30 to 60 s of rewind in games, returning the system to yesterday's state, rollback of up to 8 frames for online games.
Renewal	updates	One signed A/B image: Fortress, domains, drivers, Chameleon cores, Satellite firmware, VSA-6000 microcode. Delta downloads, written to the other half while the user works.
Chameleon	FPGA	Loads a signed core in under 300 ms and shares the CGRA between Newton, Ray and Orpheus. One retro core at a time.
Lock	privacy	Personal, Work and Public labels on files and clippings. A per-app firewall and a log of who sent what. Powerbox: a program gets access to a file only when the user chooses it.
Keeper	backups	At night or when there is surplus solar power, to a second disk, M-DISC or a cloud of the user's choice, encrypted with the Safe key.
Find	search	One index for files, attributes, mail, settings and commands. Local, with synonyms in every installed language and, for Serbian, both scripts (Latin and Cyrillic).
Datatypes	formats	The Amiga principle: a format plug-in works in every Neo program. A tree of kinds (Picture, Sound, Video, Text, Document, Table, 3D, Archive, Map); plug-ins wrap FFmpeg, libvips, libheif, libjxl, Poppler, MuPDF, LibreOffice filters (and libmwaw for old Mac formats), Assimp, libarchive and GDAL, and AROS datatypes bring the Amiga formats. Over 1,000 formats readable. Every decoder runs in its own sandbox. Multi-format clippings, live links.
Undo	history	A tree of edits inside a document, file versions through copy-on-write, system rollback through Chronos.

5. Graphics and graphics APIs

The foundation is **Vulkan 1.4** with our own driver for the VSA-6000, written inside the Mesa project (like the Turnip or Asahi drivers). Everything else is built on Vulkan, so a fix or speed-up in one place is felt everywhere. Glide is not the main API but a retro layer.

API	How it works on Neo	For whom	1.0
Vulkan 1.4	native "vsa" driver in Mesa, mesh shaders, variable-rate shading, ray tracing through Ray	new games, Neo SDK	yes
OpenGL 4.6 and **OpenGL ES 3.2**	Zink: OpenGL over Vulkan, part of Mesa	Linux programs, Blender, older games, emulators	yes
Direct3D 9, 10, 11	DXVK in Bridge	Windows games	yes
Direct3D 12	VKD3D-Proton in Bridge	newer Windows games	yes
Glide 2 and 3	Neo Glide layer on Vulkan, uses the T-Buffer for smooth edges	old 3dfx games, no wrapper needed	yes
WebGPU and **WebGL 2**	in the Horizon browser	the web	yes
OpenCL 3.0	Rusticl from Mesa, and on Newton as general compute	science, image processing	yes
Vulkan Video	AV1, HEVC, H.264 through Courier and Panorama	Cinema, Cut, Broadcast	yes
Neo 2D	Ray blitter: masks, Bézier curves, 12 Porter-Duff modes, SDF fonts, classic bitplane mode	Neo Kit, the demoscene	yes
Copernicus lists	WAIT, MOVE, SKIP, KICK… from programs, through the Neo SDK	games, the demoscene, copper effects	yes
Newton API	deterministic physics, rigid bodies, particles, fluids	games (Pinball Cosmos)	yes
SDL3, Godot, Unity, Unreal	over Vulkan; Godot and SDL3 get official support	game developers	SDL3 and Godot yes, the rest by agreement

How work is split between the graphics chips

Graphics is not one chip but four, like the 2D and 3D cards of the late nineties, but in one package and over one memory. Shading runs only on the graphics chip, because moving tiles between two separate chips would cost more than it saves.

Chip	Job	Spiritual model
Distributor (on the base die)	receives geometry, sets up triangles, splits the screen into tiles and hands them to the chiplets	3Dlabs GLINT Delta and Gamma, the setup and geometry chips
Graphics chiplets (2 to 8)	vertex and pixel shading, textures, compute, T-Buffer	3dfx VSA-100, several chips on one picture
Ray	ray tracing, blitter, 2D acceleration and layer composition	the Amiga blitter, the 2D cores of Matrox cards
Panorama	up to 6 outputs, 10 and 12 bits of colour, RAMDAC, 15 kHz, gaming across three screens, hardware-smoothed text	Matrox Parhelia: TripleHead, 10-bit colour, analogue picture quality

```
The old chips themselves (Parhelia, GLINT, VSA-100) are not used: they
are no longer made, they were built on 150 to 250 nm processes and they
```

```
have no modern shaders. The rights to them and to their names belong to
others, so we take ideas, not names.
```

Borrowed from consoles

Consoles had fixed hardware and had to get everything out of it, just like Neo. These are the solutions that carry over to Neo under our own names, and the lessons about what to avoid.

In Neo	What it does	Model
Object list in Panorama	2D games without a frame buffer: a list of objects is composed line by line, with practically no limit on sprites and no latency	Atari Jaguar, Object Processor
Order-independent transparency	the chiplets sort transparent layers per tile in hardware, so glass, smoke and foliage always look right	Dreamcast (tile-based rendering)
Distributor microcode	geometry setup is programmable, and new versions arrive through Renewal	N64, the programmable RSP
Copernicus chains	data packets travel by DMA chain straight into the chip, with no CPU involvement	PS2, DMA paths to the graphics chip
Smart cartridge	a cartridge can carry its own chip (FPGA or accelerator) over a PCIe link, for special games and tools	SNES Super FX, the big Neo Geo cartridges
Pocket	a small save card with a screen and buttons: it holds game saves and the Safe key, and works as a private second screen (cards in Midnight Poker, an inventory)	Dreamcast VMU
Neo Arcade Cabinet	the same cartridge works at home and in an arcade cabinet: kiosk mode with credits, for clubs and schools	Neo Geo MVS and AES
The previous generation inside the machine	confirms the direction of Satellite and FPGA cores: old software runs on real or faithfully recreated hardware	PS2 with the PS1 processor
Open standard	several manufacturers build Neo under licence (already in "Road to a standard")	3DO

Lessons: Saturn and Jaguar had several equal processors and poor tools, so games rarely used the hardware; that is why ARM is the single master, and the SDK and simulator are in the plan from day one. N64 had too little texture cache; that is why each chiplet has 4 MB of SRAM for tiles. GameCube shows that low memory latency is worth more than big numbers, which is the reason for the shared cache and HBM.

Compositor

The compositor speaks the Wayland protocol natively, so Linux programs (GTK, Qt, SDL, Electron) run without an X server and get Neo window frames. Our own extensions: `neo-menubar` (a global menu bar), `neo-layer` (a window gets a Panorama hardware layer, for video, games and retro screens without copying), `neo-snapshot` (freeze or restore a window through Chronos) and, new in this specification, `neo-tab` (tabs and window stacking from section 8).

6. Sound, video, voice and Neuron

Subsystem	Specification for 1.0
Orpheus sound server	A PipeWire-compatible API, so Linux programs work unchanged. A real-time channel with a 1 ms buffer. Sound objects up to 7.1.4, HRTF for headphones, room correction with the microphone, Paula mode for MOD music. An analogue synthesiser for the start-up sound.
Video	Hardware encoding and decoding of AV1, HEVC and H.264 up to 8K60. Eye cameras through libcamera, with a tuned ISP for the S, M and Pro sensors. HDMI input as a source for Broadcast.
Voice	Dictation into any field (a Whisper-class model, fine-tuned for Serbian, English and every language pack), commands translated into Lux, a reading voice (Piper class; natural Serbian, UK and US English, and a robot voice). All local on Neuron. The wake word is off until the user turns it on.
Ask Neo	A local language model of 7 to 8 billion parameters (INT4) that answers from the Library (Wikipedia, OSM, Gutenberg on the Knowledge cartridge) and always cites its source. It asks before any action that cannot be undone.
Neuron runtime	ONNX models through the IREE compiler. Shared between upscaling in games, noise removal, translation (Interpreter), OCR and voice, with priority set by the Dispatcher.

6b. Neo Sound: Hi-Fi speakers, headphones, and libraries of sound and knowledge

Good sound does not have to be loud or expensive. The idea is the same as with graphics: cheap, honest hardware, with the quality drawn out by software. Orpheus already has a DSP, room correction and HRTF. Here it gets its own speakers and headphones and knows exactly how they sound, so it can correct them better than expensive parts would on their own.

Principles

- **Low power, high quality:** the speakers are for a desk, a room and a classroom, not a party. The goal is clean, flat sound at normal volume, not decibels.
- **Wood and repair:** wooden cabinets, parts held with screws, standard drivers that can be bought separately, and spare parts and a repair guide for at least 10 years.
- **Measured, not promised:** every set has its measured response published, before and after Orpheus correction.
- **Healthy hearing:** the system keeps track of how much sound the user has listened to and protects children's hearing.

Neo Sound speakers

	Neo Sound 2.0	Neo Sound 2.1
What is in it	two small speakers	the same pair and a mini woofer
Drivers	3" full-range (paper or bamboo cone, neodymium) and a 3.5" passive radiator at the back	the same, plus a 5.25" long-throw woofer with a bass port
Cabinet	12 mm birch plywood or solid beech, internal bracing, wool damping; about 12 × 18 × 15 cm	woofer: a cube of about 22 cm, 15 mm plywood
Amplifier	class D, 2 × 15 W, in the right speaker	class D, 2 × 15 W + 30 W, all in the woofer (the speakers are passive)
Response after correction	65 Hz to 20 kHz, ±3 dB	38 Hz to 20 kHz, ±3 dB
Maximum	about 92 dB at 1 m	about 96 dB at 1 m
Distortion	under 1% at 85 dB, from 100 Hz up	under 1% at 85 dB, from 50 Hz up
Inputs	USB-C (USB Audio Class 2, 24-bit / 96 kHz, native driver), 3.5 mm analogue, optional Bluetooth LE Audio receiver	
Power draw	under 0.5 W in standby; they switch on and off with the computer over USB	
Finish	natural oak, walnut, black stain; magnetic fabric grille	
Target price	€90 to 130	€150 to 220

Build it yourself is a flat-pack kit for schools and the curious: cut panels, drivers, cables and instructions, with Neo School lessons on sound, frequencies and resonance. Once the kit is built, Orpheus measures it with the microphone and creates its own profile. The target price is about 30% below the ready-made speakers.

The sound path through Orpheus

What expensive speakers do with analogue parts, Neo does in software, on the Orpheus DSP, with no load on the processor:

1. **Room correction:** once at set-up, with the microphone in the Eye camera or the headphones, from three listening positions.
2. **Speaker profile:** every model is measured at the factory, and Orpheus corrects the frequency and phase response with an FIR filter. Autoconfig recognises Neo Sound over USB and loads the profile by itself.
3. **Crossover:** in software, between 80 and 120 Hz, with phase and delay adjusted to where the woofer stands.
4. **Loudness:** at low volume the bass and treble are lifted gently along equal-loudness curves, as on old amplifiers.
5. **Protection:** limits cone travel and coil temperature, so a small driver cannot be damaged even at full volume.

Listening mode	What it changes	Latency
Music	linear phase, full correction	about 20 ms
Games	minimum phase, spatial sound from the game	under 5 ms
Film	virtual surround, clearer dialogue	matched to the picture
Speech	for audiobooks, podcasts and lessons: emphasised voice, even volume	not critical
Night	compressed dynamics and less bass, so the house does not wake up	not critical
Pure (bit-perfect)	no processing at all, for people with their own amplifier or DAC	lowest

Music formats through Datatypes: FLAC, ALAC, WavPack, Opus, MP3, AAC, and DSD for enthusiasts; up to 192 kHz, gapless playback, album volume levelling (ReplayGain). Gramophone shows the exact sound path: which file, which sample rate, which processing, which output.

Neo Sound headphones (optional)

	Wired	Bluetooth
Sound	closed-back, 40 mm driver, 32 Ω, the same drivers in both versions and the same factory profile in Orpheus	
Microphone	in-line, with a mute switch; an optional flexible boom microphone for games and remote lessons	two elements with noise reduction, and an optional boom microphone
Cable	3 m, detachable, fabric-covered, 3.5 mm and a USB-C adapter	the same cable; the headphones work wired even when the battery is flat
Wireless	—	Bluetooth 5.4 LE Audio (LC3), SBC and AAC as fallback; two devices at once (computer and phone)
Battery	—	50 to 60 hours, USB-C charging, 10 minutes for about 5 hours, replaceable battery
Comfort and repair	about 250 g, folding, clip-on memory foam ear pads; cable and pads sold separately	
Children's version	smaller headband, volume limited to 85 dB, brighter colours	
Target price	€49 to 69	€89 to 119

Deliberately without active noise cancelling: it makes headphones more expensive and shortens battery life, and good closed ear cups are enough for a desk and a classroom. It could be a separate version later.

What the OS does for headphones: a personal HRTF (a short set-up by listening, for spatial sound in games and films), **Auracast** for sharing sound (two pairs on one computer, or a teacher and 30 headsets in a classroom), the battery level as a tile on a Shelf, sound captions when the headphones are off, and automatic switching to the speakers.

Hearing care and accessibility

- **Hearing diary:** Orpheus measures how much sound went into the headphones over the week, following the WHO recommendations for safe listening, and warns in good time. In the Kids' room the volume is limited.
- **Hearing check:** a short tone test through Neo Sound headphones gives a personal profile; Orpheus then lifts the frequencies the user hears less well. It is not a medical diagnosis, and it says so.
- **Mono and balance** for people who hear better in one ear, and captions for every system sound (section 8d).

Where else hardware can save money

Proposals for the hardware documents; the OS supports both options, so no proposal changes the software.

Proposal	What changes	Indicative saving
Speakers without their own DSP	Orpheus does the correction and crossover; one amplifier in one place, no passive crossovers	€15 to 25 per set
Professional connectors as a module	XLR, phantom power, HDMI input and 15 kHz output built in only on the Neo 4000; for the Neo 2000 an expansion card (Autoconfig)	€20 to 35 per Neo 2000
One USB-C path for sound	speakers, headphones and microphone through the same native USB Audio driver, with no separate chips	€3 to 6 per computer
5G modem and UHD Blu-ray as options	as an M.2 card and an external drive, not in every computer	€40 to 90 when left out
LE Audio instead of licensed codecs	LC3 is part of the Bluetooth standard; aptX and LDAC become extras	a smaller licence line
One microphone array	the Eye camera and room correction share the same microphone array	€2 to 4 per computer

Sound library: music and sound from the public domain

Collection	What is in it	Sources
Classical	about 300 hours of works and performances that are free: orchestral, piano, chamber and choral music	Musopen, free recordings from European broadcasters and conservatoires, IMSLP for scores
Old recordings	jazz, popular songs and folk music from 78 rpm records, only where they really are free	digitised 78 rpm collections, national sound archives
Traditional music	folk songs and dances, by country	ethnomusicology archives, with permission
New free music	contemporary artists who release under CC0 or CC BY, credited in Gramophone	Free Music Archive, ccMixter, an open call to artists through the Co-op
Retro and chip music	MOD, SID and chip music from the demoscene	only with the author's permission
For games	about 5,000 sound effects and 200 music loops, all CC0, with tags (footstep, explosion, menu, victory); part of the Neo SDK	OpenGameArt, Freesound CC0, Kenney
For school and calm	children's songs, nature sounds, music for study and relaxation	our own recordings and CC0 collections

- **Selection:** a small panel of musicians, teachers and librarians chooses recordings for the quality of the recording and the performance, not the quantity. About 50 hours are on the Knowledge cartridge, the rest in Treasury.

- **Data:** every recording carries the performer, work, year, source and licence (MusicBrainz tags), so Gramophone always says where a sound came from and who made it.
- **Rights per country:** the public domain does not last the same everywhere (for example 70 years after the author's death in the EU, different elsewhere, and a recording and the work it records have separate rights). Treasury shows only what is free in the user's country.

Audiobooks

- **Ready-made recordings:** LibriVox and national collections of public-domain recordings, with a per-country rights check.
- **Read to me:** any book in the Library (Gutenberg, Wikisource, national digital libraries) is turned locally into an audiobook with a voice running on Neuron, with no internet, with a choice of voice and speed.
- **Family library:** a grandparent or parent records themselves reading a book or telling a story, and Neo keeps it as an audiobook with chapters, only in Home, labelled Personal.
- **Audiobook mode** in Leaf and Gramophone: bookmarks, speed without changing the pitch of the voice, a sleep timer, following the text while listening (good for learning to read), and carrying on from the phone through Tether.
- **DAISY** and Braille display reading for blind and partially sighted readers, together with the screen reader.

Libraries of knowledge and local versions

The libraries are part of the **country pack** (section 18b): one signed pack per country or language, containing only data, never code.

Part of the pack	Example for Serbian	Example for English
Encyclopaedia and dictionary	Wikipedia and Wiktionary in Serbian, both scripts	Wikipedia, Wiktionary, Simple English Wikipedia for children
Books	Wikisource, free works from the National Library of Serbia's digital collection (for example Njegoš, Vuk Karadžić)	Project Gutenberg, Wikisource
Audiobooks	Read to me with a Serbian voice, family library	LibriVox (a large catalogue), Read to me with a British or American voice
Music	folk music from archives, with permission	free recordings of British and American traditional music
School	textbooks and lessons where the country grants the rights	open textbooks (for example OpenStax)

The same applies to every language: a country or community adds its own sources to the country pack, and the Neo team checks rights and quality.

Bundles and prices

- **Neo 2000 + Neo Sound 2.1:** a home bundle, at a discount compared with buying separately.
- **Classroom with sound:** 30 children's or standard wired headsets and Auracast for the teacher, with Neo for schools (section 18b).
- **Build it yourself** for design and technology lessons, with lessons in Neo School.

```
Software work: about 95 P-M (speaker and headphone profiles, listening
modes and pure output, LE Audio and Auracast, hearing diary and hearing
check, sound library and per-country rights, audiobooks, Read to me and
the family library, country pack framework), counted under Media and
voice and under Tools and ecosystem. Designing and manufacturing the
```

```
speakers and headphones is not in the software budget; every sound
figure is a target to be measured on a prototype.
```

7. Files, packages, updates and security

Places in the system

Logical name	Where it lives	Who changes it
`System:`	EROFS image in pSLC ROM (A/B), LZ4 and zstd, unpacked in hardware by Courier	only Renewal
`Apps:`	neopak files: one signed, compressed file per program, mounted without installation	Treasury and the user
`Home:` (`Documents:`, `Pictures:`, `Games:`, `Pads:`)	mirrored Btrfs on two M.2 drives, attributes and a query index	the user
`Settings:`	readable text files, one folder per program	the user and the program
`Boxes:`	Linux, Wine prefixes, Amiga `DH0:`, floppy images	each box only its own folder
`RAM:`	RAM disk, emptied at shutdown	everyone
`RAD:`	a persistent RAM disk that survives a warm restart and a change of FPGA core (described below)	the user

RAD: the RAM disk that survives a restart

On the Amiga, the RAD: disk stayed in place after a warm reset (Ctrl + Amiga + Amiga), so programs and files loaded instantly. NeoOS brings it back under the same name, `RAD:`, which is a generic abbreviation and not a protected name.

- **How it survives:** on a warm restart the boot loader in NOR does not clear the reserved part of memory, and Fortress mounts it again. Every block has a checksum; a damaged disk is not mounted and the user gets a message explaining why.
- **What it survives:** a warm restart, a change of FPGA core in Chameleon, a restart after a Renewal update and a restart after a system crash. It does not survive a power cut, except on the Neo 500 while the battery lasts.
- **Size:** at most 4 GB on the Neo 500, 16 GB on the Neo 2000 and 32 GB on the Neo 4000; it only takes as much memory as its files need.
- **Security:** the contents are encrypted like the rest of RAM, and Safe keeps the key only across a warm restart. Lock labels apply here too.
- **Booting from it:** as on the Amiga, the system can start from `RAD:`, which developers use to try out test images and FPGA cores quickly.
- **At shutdown:** the option "Keep RAD: at shutdown" writes the contents to the SSD through Courier (compressed, in a couple of seconds) and restores them at the next power-on.

Memory without swap

The rule: NeoOS never writes memory to disk page by page. Everything that matters for speed stays in RAM, and whatever must survive a restart lives in `RAD:`. There is enough memory for this: 32 GB on the Neo 500, 64 GB on the Neo 2000 and 96 GB on the Neo 4000.

Layer	What it holds	Survives a warm restart
System in RAM	the whole `System:` image unpacked into RAM at power-on (on the 500, only the most used parts)	not needed, it reloads from ROM in under a second

Layer	What it holds	Survives a warm restart
`RAD:` fast layer	the programs the user uses most, their libraries and caches, with pre-linked code; the Dispatcher chooses what goes in from the user's habits	yes
`RAD:` live documents	every open document has a live copy with its undo history; the SSD gets a copy in the background through Courier	yes, the document comes back with the cursor in the same place
`RAM:`	a dynamic temporary disk that grows and shrinks as needed	no
Program working memory	what a program is using right now; optionally compressed in RAM (LZ4 through Courier), never to disk	through a Chronos snapshot

- **When memory runs short**, the Dispatcher works in this order: caches, the `RAD:` fast layer, program code that can always be re-read from ROM or a neopak package, background memory compression (Courier), putting a whole background program to sleep with a Chronos snapshot, and finally a message telling the user what to close. The program in the foreground is never touched.
- **Power cut:** on the Neo 2000 and 4000 an on-board ultracapacitor gives a few seconds of runtime, enough for Courier to write a compressed copy of `RAD:` to the SSD; on the Neo 500 the battery does this. Everything comes back at the next power-on. ECC memory and periodic scrubbing protect the contents while the computer runs.
- **How much faster it is:** RAM is about 100 times faster than an NVMe drive, but the system already keeps recently used files in cache, so the gain is not a hundredfold. Estimate: the first launch of a program after power-on goes from about 300 to under 100 ms, opening a document is practically instant, and getting back to work after a warm restart drops from about 10 s to about 2 s. In work that is already "warm" the difference is small, about 10 to 20%. It is felt most after power-on and restarts, and on the Neo 500 with its slower drive.

Security in 1.0

- Secure boot: the NOR boot loader checks the image's signature with the Safe key, and only then starts Fortress. An unsigned image does not boot, and service mode over USB-B restores the factory image.
- ARM MTE for Rust and C++ services, an IOMMU for every device, encrypted RAM; Satellite and PCIe cards see only the memory windows assigned to them over CXL.
- Physical switches for the camera, microphone and network, with an LED that software cannot turn off.
- Treasury: reproducible builds, and the system checks that an installed program matches its published source code.
- Error messages with a code (for example `NEO-FS-017`) that say what happened, why, and what the user can do.

7b. Neo card, Autoconfig and portability

The computer is upgraded by swapping the **Neo card**: it carries the Neo package (an ARM compute chiplet with 4+4 to 8+4 cores, 2 to 8 graphics chiplets, the helper chips, I/O with the Distributor, FPGA) and a ROM describing the card. The Phase 1 card carries an off-the-shelf ARM SoC and an FPGA. The same slot stays valid for at least 8 years: a new card works in an old board and the other way round. There are no hardware hyper-threads.

Requirement	What the OS does
Recognising the card	The boot loader reads the description from the card's ROM (cores, chips, memory, version) and picks the matching native drivers from the system image. Swarm and the Dispatcher take the core count and chip list from the description and relearn what runs faster where.
One image for every card	Renewal ships one image with drivers for all supported cards; a new card in an old board boots without reinstalling, with the same Home and RAD: disk.
Autoconfig	Modelled on Amiga Zorro: every expansion card carries a description, an icon, a signed driver and help. The driver runs in its own Fortress cell behind the IOMMU, so a bad driver cannot endanger the system. The card appears in Cabinet and in the Caretaker by itself.
Portability to RISC-V	The third-generation compute chiplet may be RISC-V. seL4 and Genode already support it; the Rust services and Neo Kit have no ARM-specific code except in a thin platform layer. From day one, CI builds and tests a RISC-V version in QEMU. Hand-written SVE2 assembly has an RVV counterpart or a pure Rust fallback.
Under the bonnet	shows which card is fitted, its temperature, power draw, ROM version and the state of every chiplet.
Pin header and user FPGA area	GPIO, I²C, SPI and UART on the board and a part of the FPGA reserved for the user are available from Lux and Neo BASIC, behind Keys, like the user port on old home computers.

7c. Neo 100, Neo Revive and Neo OS Lite

Neo 100 is a small computer inside a keyboard (ARM board, 4 to 8 GB, no FPGA, VSA-6000 or Satellite, about €60 to 90). **Neo Revive** runs the system from a USB stick on old x86 and ARM computers. Both use **Neo OS Lite**.

Neo OS Lite has	Lacks or changes
the same Fortress, Neo Kit, Datatypes, Quartet, Leaf, Board, Lux and Neo BASIC	only the Retro style and the Calm level
Treasury as a signed mirror on the stick	emulators instead of FPGA cores
Co-op and the Guest room for sending heavy jobs to a bigger Neo in the Neighbourhood	apps with a 1 to 2 GB RAM budget
Neighbourhood mail over direct Wi-Fi	no RAD: survival on hardware that cannot support it

x86 support in Revive: the exception to the driver rule

For the Neo 100 the rule from section 3 applies: it is our hardware and it has native drivers. For Revive that is impossible, because old PCs contain thousands of different chips. So Revive uses Linux driver code in isolated components (the same Genode method as in section 12b), with a list of verified computers:

- **x86:** 64-bit Intel and AMD computers with UEFI, roughly from 2012 on, with at least 2 GB of RAM; Intel and AMD graphics through Mesa, Nvidia only basic (nouveau). seL4 and Genode already run on x86_64. 32-bit x86 is not supported.
- **ARM:** a few popular boards (Raspberry Pi 4 and 5 class), not every ARM device.
- **The "Does it work?" database** records the models users have checked; an unknown computer first runs a 2-minute test.

Lite on phones

Possible, but as an experimental option for 2.0. Phone parts (modem, cameras, battery management) mostly have closed drivers, so Lite runs only on phones with open support (PinePhone class, Librem 5, Fairphone and a few older Android models from the postmarketOS list), using the same Revive driver method, with the touch mode from section 8h and calls through ModemManager. Expected result: good for reading, messages, mail and Quartet, weaker battery life and camera than on Android. Estimate: about 200 person-months, version 2.0.

When and how much

Part	P-M	Version
Neo OS Lite profile (memory budgets, Retro and Calm, emulators, Treasury mirror)	50	1.1
Neo 100: native drivers for the board	60	1.1
Revive for x86: platform layer, Linux driver components, a lab with 60 to 80 old computers	150	1.5
Revive for ARM boards	40	1.5
Total after 1.0	**300**	**about 12 months, 20 to 25 people from existing teams**

7d. Crunching: programs and data on the fly

On the Amiga, programs such as Turbo Imploder and PowerPacker "crunched" executables: a program took half the space on a floppy and unpacked itself as it loaded. NeoOS brings the idea back, but as a service of the system rather than a trick inside the program. It applies to the full NeoOS and to Neo OS Lite alike.

Does it still make sense

Where	What you gain
Neo 500, 2000, 4000 (NVMe, Neo ROM)	mostly space: programs and documents take about 30 to 50% less. Loading changes little because NVMe is already fast, but nothing is lost: **Courier decompresses in hardware** (part 3), with no processor work.
Neo 100 and Neo Revive (SD card, eMMC, USB stick, old SATA disk)	space and speed: when the disk is slower than decompression, fewer bytes read means a faster start. On an old disk or a USB 2.0 stick a program can load up to twice as fast.
Treasury and Renewal	smaller downloads, which matters most on a slow or expensive connection and for country packs.

```
The figures are estimates from published measurements of zstd and LZ4 on
typical executables and documents; the lab measures the real values on
all three models and on the Revive computer list.
```

Crunched programs

- **No unpacker inside the program.** Imploder and UPX put a small unpacker at the start of a program, so the program rewrites itself when it starts; antivirus tools often flag that falsely, and the memory cannot be shared. On Neo a program is an ordinary ELF file and the **system loader** (part 4, Exec-N) decompresses it on the fly.
- **In 64 KB chunks**, like the EROFS image: only the parts of a program that are actually used are loaded, and a clean code page can be dropped from memory and read again from disk as if it were not crunched.
- **The signature stays valid.** The program's content is what gets signed, and each chunk has its own checksum, so Safe verifies a program whether it is crunched or not.
- **When:** Treasury ships `neopak` packages already crunched (part 7). Programs from other sources are crunched at installation or later, in the background, while the computer is on mains power and idle.
- **Dictionaries for small programs:** tiles, Helpers and Neo DOS commands are small, so they are crunched with a shared zstd dictionary built from system code. Even a 40 KB program gets a decent saving that way.
- **Linux programs** in the Applications domain are left as they are: the file system underneath crunches them.

Data crunched on the fly

For documents and data, compression is built into the file system, so no program sees it: programs read and write ordinary files, and on disk they are stored crunched. The Storage domain uses Btrfs (part 3), which can already do this.

Place	Algorithm	Why
`System:` (ROM)	EROFS with LZ4 and zstd	already so in part 7; Courier decompresses
`Home:` and other disks, full NeoOS	**zstd**, level 3 by default	a good balance of saving and speed; with Courier, decompression outruns NVMe
`Home:` on Neo OS Lite	**zstd**, level 1	the fastest zstd level, close to LZ4 in speed, easy on the battery and a weak processor (Btrfs does not offer LZ4, so this is its fastest strong choice)

Place	Algorithm	Why
`RAM:`, `RAD:`, compressed working memory	**LZ4**	only speed matters; the same as in part 7
Archives and long-term keeping (Archive, Keeper, M-DISC)	**xz** or **zstd** at level 19 and above	the biggest saving; packing is slow but rare

- **A level per file and folder:** in the Info window ("Crunch: no, fast, strong, strongest") and with the Neo DOS command `Crunch Home:Archive LEVEL 19 ALL`; `Decrunch` undoes it.
- **No wasted time:** files that are already compressed (JPEG, video, MP3, ZIP) are recognised by their first bytes and left as they are.
- **A copy to a USB stick or a mail attachment** always goes out as an ordinary file, readable on any computer.

Why not RAR

RAR is a closed format: its author lets other programs only **extract** it, and packing into RAR is reserved to the author's own software. So RAR cannot be the packing format of an open system. Little is lost: **xz** and **zstd** at their highest levels save space in the same class as RAR5, and they are open and quick to unpack.

- **RAR archives open** through libarchive (open source, RAR 4 and most of RAR5), in a sandbox like every decoder (part 4, Datatypes). The official unrar is an add-on in Treasury for the rare archives libarchive cannot read.
- **A recovery record**, RAR's best feature (repairing a damaged archive), comes from open PAR2 data stored next to the archive: Archive and Keeper add it by themselves, for M-DISC and USB sticks.
- Packing into ZIP and 7z (7-Zip, LGPL) for exchange with other computers.

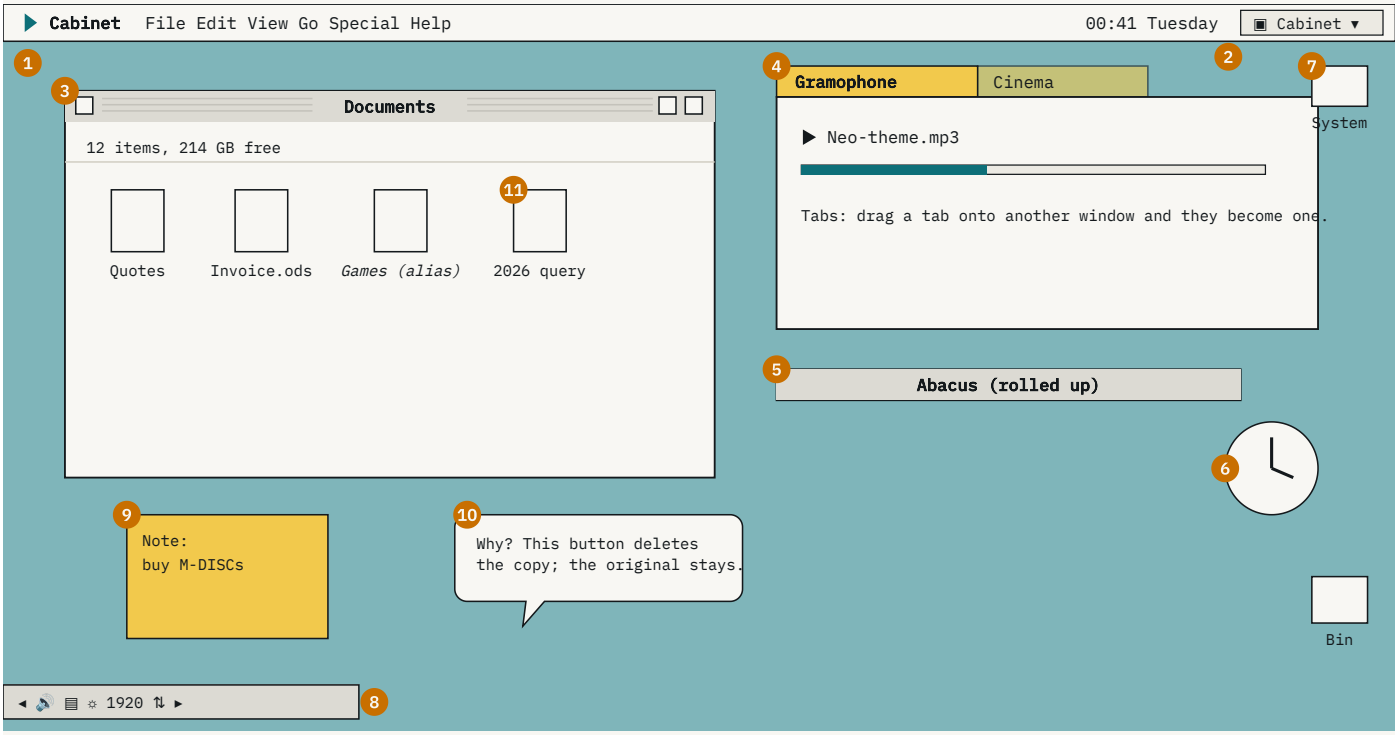
The Disk tile and the "Crunch" button

- The **Disk tile** (part 8i): the small face shows free space, the large face also shows what crunching has saved ("Saved 38 GB").
- The **"Crunch" button**, a nod to Imploder: it re-crunches larger folders at a stronger level, in the background, while the computer is on mains power (Dispatcher, class 3). Progress in the Retro style shows calm colour bars in the spirit of Imploder's stripes, with no flashing.
- **In the Lite version** the same button and tile; the default level is lower, and stronger crunching runs only on mains power.

```
Estimate: crunched programs in the loader, dictionaries, the compression
policy in Storage, the Crunch and Decrunch commands, PAR2 with archives,
the Disk tile: 15 person-months. Lite uses the same code.
```

8. Look: Neo Silver

The guideline: as many elements of the classic Mac OS environment (System 7 to Mac OS 9) as possible, with the best of BeOS, Haiku and MorphOS. Everything is our own drawing: the Neo mark instead of an apple, our own icons, our own font and our own names. We take the behaviour and the spirit, not the graphics or the trademarks.



1 Neo menu and global bar · 2 program menu · 3 Silver window, spatial Cabinet · 4 Haiku tabs · 5 window shade · 6 replicant · 7 disks on the right, Bin at the bottom · 8 shelf with live tiles · 9 note · 10 "Why?" balloon · 11 live query as a folder

The elements and where they come from

Element	Model	In NeoOS
Classic Mac OS (the main character)		
Global menu bar at the top	System 1 to Mac OS 9	One bar for every program, including Linux and Windows windows through `neo-menubar` and Link. The screen edge is an infinitely large target for the mouse.
Neo menu on the left	Apple menu	The Neo mark opens: About This Computer, small programs, recent files, the user's own additions (folder `Settings:Neo menu`).
Caretaker on the right of the menu bar	Application menu, 7.5 to 9	The list of open programs (who is doing what in the house), Hide Others, and it can be torn off into a floating strip. Launching lives on the Shelves and what is running lives here, so the two never mix.
Spatial Cabinet	the classic Finder	Every folder is one window that remembers its position, size and view. Two panes are another mode of the same Cabinet, one key away.
Disks on the right, Bin bottom right	the Mac OS desktop	System:, Home:, the cartridge and USB as icons on the desktop. Eject by dragging to the Bin.
Silver frames	Mac OS 8 and 9	Light grey, bevelled buttons, a striped title bar, close, zoom and shade buttons.

Element	Model	In NeoOS
Window shade and spring-loaded folders	Mac OS 8	Double-clicking the title rolls the window up; a folder opens by itself while you hold a file over it.
Shelves	Control Strip, Amiga dock programs	Instead of a Start menu and a Dock: as many shelves as you like, along any edge or free on the Desk. Anything dragged onto them goes there (programs, documents, folders, Lux scripts), plus live tiles (clock, mixer, RAD:, Sun). Every task has its own shelves. The command palette covers the rest. Neo arrives with one ready-made shelf at the bottom.
Helpers	Amiga Commodities	Small background tools that each do one thing (hotkeys, focus follows mouse, abbreviations, automatic actions), written as Lux add-ons, with one panel to switch them on. They get keyboard and mouse access only through Keys, so they cannot become keyloggers.
Control panels	System 7	Every setting is a small separate window with a "Why?" button. Search joins them together when needed.
Help balloons	Balloon Help	Switched on from the Help menu; every balloon is "Why?" text in plain language.
Label colours	Labels	Seven colours with names the user can change ("Urgent", "To print"), also usable in queries.
Info window	Get Info	Comment, size, version, Lock label and every attribute of the file (in the Haiku spirit).
Aliases in italics	System 7	A shortcut still works when the original is moved.
Small programs	desk accessories	Notes (sticky notes), Calculator, Puzzle, Keyboard Map, Scrapbook (clippings).
About This Computer	About This Mac	Memory bars per program, and which part it runs on (ARM, Satellite, FPGA core).
Icon parade	extension icons at start-up	On a cold start, icons of the loaded parts and the FPGA core march along the bottom of the screen. It lasts as long as start-up, about a second.
Themes and sound sets	Appearance	Silver, Slate, Contrast, Phosphor; the "Neo" sound set from the sound pack.
Easy mode	Simple Finder, At Ease	An Easy/Pro switch for the whole desktop, for children and beginners.
System font	Chicago, Charcoal	Our own "Neo System" font in the spirit of bitmap fonts, sharp at every size through SDF, covering Latin, Cyrillic and Greek.
BeOS and Haiku		
Window tabs	the BeOS yellow tab	The title is a tab that slides along the top edge. Drag a tab onto another window and they become one tabbed window.
Stacking and tiling	Haiku Stack & Tile	Windows glued edge to edge move together; the layout is saved as the task's workspace.
Replicants	BeOS	A part of a program (a clock, Gramophone controls, a CPU graph, the weather forecast) is dragged onto the desktop and lives there.
Live queries	BeOS Tracker	A search is saved as a folder that refreshes itself; attribute columns in Cabinet.
Workspaces	BeOS Workspaces	Up to 9, each with its own resolution, background and even a retro core.
A thread per window	BeAPI	No stuck window blocks the others, nor the menu bar.
Vector icons	Haiku HVIF	Small, sharp at every size, with Silver-style bevels.
MorphOS and Amiga		
Drag-down screens	Amiga, MorphOS	Pull the desktop down and underneath is a retro core or the Windows layer.

Element	Model	In NeoOS
Per-program appearance	MUI and Zune settings	Each program can have its own text size, density and colours without breaking the look of the whole.
Instant response and smooth animation	Ambient	Animations last at most 150 ms and can be switched off entirely (levels in section 8d). Draw first, animate after.
Right button for menus	Amiga	Optional: hold the right button and the menu bar appears anywhere, as on the Amiga.

> **Naming rule:** NeoOS does not use anyone else's names or licensed names. The "Model" column only cites the historical source of an idea; everything the user sees (themes, programs, font, sounds, icons) carries our own name and our own drawing. That is why the theme is "Neo Silver" and not Platinum, and the dark variant is "Slate", not Graphite. Before the beta, a lawyer checks every name and the whole look.

8b. Working with tasks and documents

Lisa (1983) started from the idea that the user works with a document, not with a program: you do not "launch Abacus", you tear a new sheet off a pad of spreadsheets. NeoOS takes this as a basic rule and adds related ideas from other systems that thought the same way. Programs still exist, but they stay in the background: the system shows tasks and documents.

Idea	Where from	In NeoOS
Pads instead of "New"	Lisa, 1983	`Home:Pads` holds the pads (Letter, Invoice, Household budget, School essay). A double-click tears off a new sheet with a name and date, already in the right place. Anyone can make a pad from one of their own documents.
No "Save", no "Quit"	Lisa	A document saves itself, and closing it leaves it exactly as it was, with the cursor and history. A program closes by itself when none of its documents is open.
Set aside	Lisa "Set Aside"	A document shrinks to an icon on the desktop and stays "open" without taking up the screen.
The Desk comes back	Lisa, the soft power button	Switching off (Nap or Good night) remembers every open document, window and tile. At the next start-up the Desk is exactly as it was, through Chronos and RAD:.
A disk carries its documents	Lisa	When a cartridge, USB stick or disk is taken out, its open documents save themselves and close neatly, with no error message. When the disk comes back, the documents reopen in the same places.
Window and icon are one	Lisa, the classic Finder	While a document is open its icon is hollow; double-clicking it brings the window to the front. A closed window "shrinks" back into its icon, and the document remembers its window's size and place.
Tasks as rooms	Xerox Rooms (1986), Magic Cap (1994)	A **task** is a named whole ("Tax return 2027", "Dissertation"): documents, windows and their layout, notes, a deadline and a Lock label. Switching to a task brings everything back as it was left, through Chronos. A finished task is archived as one Case.
The same commands for every object	Xerox Star (1981)	The keyboard's function row has Move, Copy, Properties, Again and Undo keys. They work the same on a file, a sentence, a cell, a picture and a window. Properties opens one property sheet for any selected object.
Jump	Canon Cat and SwyftCard (Raskin, 1987)	Hold the Jump key and type: the cursor leaps straight to that text, forwards or backwards, in any document or list. Faster than the mouse and scroll bars.
Compound documents	OpenDoc (1996), OLE	A document is a set of parts: text, table, picture, sound, map. Each part is edited in place, in the same window, with the help of Datatypes. A part from another document can be a live link.
Data without files	Newton "soups" (1993)	Notes, contacts, reminders and tasks live in one store, so "Call Mark tomorrow" written anywhere becomes a reminder in Diary and an item in a task.
Help that understands intent	Newton Assist	Select some text and press Voice or Ask Neo: "send this to Ana", "make a table". It is translated into Lux commands and asks before sending.
Text as a command	Oberon (Wirth, 1987)	Any Lux line in any text (a note, an email, documentation) runs with a middle click. The instructions are also the button.
Object first, then action	Xerox Star, Smalltalk	First you select the thing, then you see what can be done with it (Services from the menu, right-click). The same order everywhere.

A retro theme in the Lisa spirit: "First Desk"

Lisa's principles are in the standard Desk for everyone. Its look is a separate theme for the Retro style, under our own name and with our own drawings (the Lisa name, its fonts and its icons belong to others):

- **Black and white**, one bit per pixel, with dot patterns instead of grey tones for the background, bars and shadows.
- **Our own bitmap font and icons** in the spirit of the early 1980s, Latin and Cyrillic.
- **A "mechanical" sound set:** relay clicks, disk whirr and a short beep, recorded for Neo (CC0), as a fourth sound theme next to Crystal, Wood and Chip.
- **Non-square pixels if you like:** Panorama shows the Desk at 720 × 364 with pixels taller than they are wide, as on the original, or with square pixels for sharpness.
- It works in Neo OS Lite too, as the lightest theme of all.

What it looks like at work

1. The user opens the task "Tax return 2027" from the Tasks menu (in the menu bar, next to the Neo menu).
2. The income spreadsheet, the PDF form and a note come back, in the same places. The deadline is in Diary.
3. From the "Letter to the tax office" pad they tear off a new sheet and drag part of the spreadsheet into it as a live link.
4. Jump key, "VAT": the cursor is on the right row. Nothing is saved by hand.
5. When done, "Archive task": everything goes into one Case with the Lock label Personal.

```
For 1.0: Pads, automatic saving, the Desk coming back, documents that
follow their disk, the First Desk theme, Tasks, the universal keys,
Jump, the Newton-style store and text as a command. In 1.0 compound
documents work inside Quartet and Diary, and as a general mechanism for
all programs in 1.5.
```

8c. Personality: everyday language and Dijana

The goal is a system you learn in one afternoon, that is pleasant and that gives results quickly. The lesson from the past: in 1995 Microsoft Bob turned the whole system into a cartoon house and failed, and Clippy annoyed people because it interrupted their work. So the metaphor lives only in the words, and only where it makes things clearer, and the guide never imposes itself.

Everyday language

In Easy mode the user sees everyday names; in Pro mode they see the technical name. Scripts and documentation use the technical names, so they work in both modes. Every language pack chooses its own everyday names; below are the English ones.

Easy	Pro (technical)	Why this name
Desk	desktop	where you work and leave things
Folder	folder	papers in a folder
Letterbox (UK) / **Mailbox** (US)	downloads	everything that arrives from outside
Bin (UK) / **Trash** (US)	deleted items	things can come back until it is emptied
Foundation	`System:`	not to be touched, everything stands on it
Toolshed	`Apps:`	tools for the job
Guest rooms	`Boxes:`	Windows and other guests, each in its own room
Notepad	`RAM:`	scrap paper for quick notes, thrown away at the end of the day
Worktable	`RAD:`	stays as you left it
Caretaker	Dispatcher and the list of open programs	knows who is doing what in the house
Keys	permissions (Lock powerbox)	a program gets a key only from you
Neighbourhood and **World**	local network and internet	near and far
Household and **Kids' room**	user profiles, child profile	who lives on this computer
Back to yesterday	Chronos system rollback	says exactly what it does
Nap and **Good night**	sleep and shut down	from everyday speech

The Lisa pads from section 8b (`Pads:`) are called **Templates** in Easy mode, so they are not confused with the Notepad for `RAM:`.

Dijana, the resident and a guide who does not get in the way

Behaviour	How
Never interrupts	She sleeps in a corner of the menu bar. When she has a useful tip, she only pricks up her ears; you click if you want to.
Speaks up on her own only three times	at the first power-on, after an error (to explain what happened, with the NEO code) and when you call her by voice or with a key
Learns	She gives each tip once. "I don't need this" switches off that whole kind of tip, and after being ignored three times she goes quiet by herself.

Behaviour	How
Sniffs things out	Search is "Dijana, sniff out...": while Find works she sniffs, and for a lost file she says where it is and when it was last seen.
Jokes, rarely	When you drag a document to the Bin she sometimes tries to "eat" it, at most once a day; one click and she gives it back. The joke has an off switch.
Knows the context	through Lux she knows what is on screen, answers locally from the Library and never sells anything
Adapts	More present in the Kids' room and Easy mode, almost invisible for experienced users, off with one switch. Not animated on battery.

The system's personality

- **A face for the system**, like the smiling computer on the early Mac: at power-on Dijana wakes up and pricks up her ears; on an error she is a sad Dijana with a NEO code and a fix; at shutdown she says "Good night" and curls up.
- **Dijana's day**, a living screen saver: a serial story that follows the real time of day, the weather outside and the calendar (holidays, seasons). Neuron assembles the scenes from ready-made parts, so they rarely repeat. Demoscene effects sit alongside it.
- **Small happenings on the Desk**, rare and switchable off: Dijana sometimes takes a walk along a shelf. Never across a document, and never while a game or video is running.
- **Sounds** from the Neo sound set, quiet and short, with one switch.

The system's voice and motion

- **Form of address:** in English the system speaks in a friendly, plain second person; in Pro mode and business profiles the tone is more formal. In Serbian it uses the informal "ti", and the formal "Vi" in Pro mode and business profiles. Each language pack sets its own rule.
- **Neo greeting:** a sound mark of the notes C-E-G-C-E in 240 ms, synthesised in Orpheus.
- **Motion curve** for every animation: `cubic-bezier(.2,.9,.25,1.15)`, with a gentle bounce at the end.
- The other identity rules (colours, type, tone of writing) are in a separate identity guide, alongside the overview.

Residents

Dijana is the first resident, but not the only one. Residents are optional mascots from Treasury and the system's only commercial add-on.

- **Bought once**, with no subscription, and most of the income goes to the author. Free community residents come under a free licence.
- **Well-known characters** come in only with an official licence. The first candidate is the snake from the song "Ide zmija" (DJ Nick DeeTronic), as a dancing resident for Gramophone. No unofficial copies and no real people; a country can add a national character.
- **The same set of behaviours** as Dijana: sleeps, sniffs, jokes, is happy and sad, has a screen saver story and its own sounds. A resident is made with a tool from the Neo SDK, as a package of animations and rules.
- **Safe by design:** a resident only receives events from the system. It sees no files and no network, and carries no advertising.

Personal touches

The computer's name and greeting, one accent colour (with a contrast check), a background, a start-up sound (ready-made or composed in the Orpheus synthesiser), a choice of voice and how

chatty it is, a **Nook** on the Desk (a picture, a quotation and a Pot plant that grows and never wilts) and the computer's anniversary with a local review of the year.

Simplicity in the style of mobile apps

- **"What are you doing today?"** is the start screen: large tiles with the tasks from section 8b and recent documents.
- **First steps:** a 15-minute guided tour at the first power-on, and every program shows at most three things the first time.
- One main action per screen, clearly highlighted; the rest is in Pro mode.
- A search field at the top of every app with more than one screen of content.
- Instead of "Are you sure?", the action happens at once and a bar appears at the bottom: "Deleted. Undo".
- No Save button; empty screens offer "Start from..." with an example.
- The same gestures everywhere: swipe to delete or archive in lists, pinch to zoom, long press for properties.
- **The yardstick:** a new user sends an email, writes a letter and finds a photo within the first 10 minutes, without a manual. This is a lab test before 1.0.

8d. Beauty, sound, files and the Co-op

Areas agreed alongside the overview and the technical book (chapters 2.2c to 2.2h), shown here with what they require from the OS.

Beauty without the weight

- **Glass** (translucent surfaces with blur) is drawn by Ray, with an automatic text contrast check.
- **Live tiles** are Panorama layers and refresh at most once a second, so they cost no CPU.
- **Newton physics** for windows that sway gently, the Bin and the seasons on the Desk.
- **Flip through**: a 3D overview of every window with its name, filtering by typing and a return to the same place.
- **Calm, Lively and Lavish levels.** Animations last at most 150 ms; on battery the level drops by one.
- **"Under the bonnet"** in the Caretaker shows what each chip is doing; **"Show what I can do"** is a demo program with a Shop window mode for shops and trade fairs.

Sound

- Under 5 ms from event to sound (the Orpheus buffer is 1 ms).
- Sound comes from the place on screen where the event happens and follows the physics, and music dips by itself when a system sound arrives.
- A caption for every sound, for deaf and hard-of-hearing users and for quiet rooms.
- Sound themes of up to 40 events in an open format (Opus or FLAC plus a list), in Treasury alongside Residents. Neo Basic, Crystal, Wood and Chip are free.

Files in Easy mode

- No Save: the name is taken from the first line and can be changed at any time.
- A file lives in one place; everything else is a link, so nothing is duplicated.
- Views by time and by person, piles on the Desk, the **Case** (everything about one thing together) and the **Bag** (what I am taking to another computer or a USB stick).
- **Cabinet Pro** for experienced users: lists, button strips, Lux scripts and modules, in the spirit of the best Amiga file managers.

Co-op

- **Borrowing power:** computers in the Neighbourhood share processor time in a closed room, and every result is checked twice.
- Reproducible program builds and a shared build cache, with a "See how it works" button in every program.
- **Clubs without a centre** and the **Board**, a built-in BBS: "Share with the club" in every app.
- **Neighbourhood mail**: messages travel over the internet, Wi-Fi, LoRa radio or on a cartridge.
- Code from a club runs in a sandbox until two members have signed it.

"Board" is the name of the built-in BBS; in Easy mode `RAM:` is called the Notepad (section 8c).

8e. Two styles and ideas from the Unix world

Retro and Future Shock

The system has two global styles. Switching takes one click, needs no restart and applies to the whole system.

	Retro	Future Shock
Look	clean, in the spirit of classic Mac OS, no effects	every effect from Beauty without the weight (section 8d): glass, physics, Flip through
Type	our own screen typeface "Pixel Block"	Neo System, sharp at every size
Icons	32×32, pixel art	vector, with bevels
Dijana	small and made of pixels	full animation
Calm, Lively, Lavish levels	do not exist	yes

- **No extra code:** every Neo Kit app works in both styles, because the style is drawn by Neo Kit and Panorama, not by the app.
- **Exceptions:** a style can be set separately for one app. On battery, Neo suggests Retro once.
- **Colours** (Silver, Slate, Contrast, Phosphor) are independent of the style.
- **Treasury** checks every app in both styles, and Neo Studio shows them side by side.

From the Unix world

The best habits from macOS, NeXT and the Linux world, under our own names.

In Neo	What it does
Space to preview	a selected file opens in a preview when you press the space bar, through Datatypes
Services	selected text or a picture from any program goes to another program (translate, send, convert)
Recipes	ready-made chains of actions (Lux) that the user assembles by dragging and shares
Carry on	work started on one Neo computer continues on another in the Neighbourhood
Pass to a neighbour	send a file to a nearby computer or phone, with no account and no internet
One mouse	one mouse and keyboard for several computers on the desk
Move house	transfer all data and settings from an old computer (including from Windows)
Disk mode	over USB4, Neo acts as an external disk for another computer, for data rescue
Zero-setup Neighbourhood	printers, shared folders and other computers appear by themselves (mDNS and DNS-SD)
Driverless printing	modern printers work at once, through IPP Everywhere
File clones	a copy of a large file is instant and takes no space until it is changed (Btrfs reflink)
Dual packages	one neopak holds an ARM and an x86 version; the Dispatcher chooses
Magnifier	zoom in on part of the screen with one shortcut, alongside accessibility
Portals	programs in the Linux and Android rooms ask for files and devices through the Neo dialogue and Keys
Firmware through Renewal	peripheral firmware arrives with the system, never separately
Neo in your pocket	the whole system on a USB drive or cartridge, runs without installation, with its own RAD: and Home

8f. Even easier

Area	What the OS does
Input to suit the job	Keyboard for text; touch in Easy mode enlarges targets by itself; a pen for writing and drawing; voice only while the "Speak" key is held (local, with an LED); the trackpad gives haptic feedback through Newton; a gamepad drives the whole system; NFC "Card tap" as a Helper (a card opens a task or signs in a household member).
Guest USB	An unknown device first goes into an isolated room (the Peripherals domain); a device that claims to be a keyboard must type a code shown on screen, so a fake keyboard cannot type commands.
Call your own	Remote help from a household member or friend, end-to-end encrypted: viewing only at first, control only after a "yes", it switches itself off after 30 minutes, and everything labelled Personal is blurred.
Record what I do	A series of actions is recorded as a Helper or a Lux script and repeated with one click.
Progress	Part of Neo Kit: Swarm measures how much is left, display by stages, "Stop" and "Carry on later" (survives a restart through RAD:), designs for Retro, Future Shock and Dijana; shown in the notification and on the icon, and without animation at the Calm level.
Shared parts	Modelled on the Enhancer pack for AmigaOS 4: toolbars that overflow into a menu, tabs that can be dragged out into a window, a pie chart, media controls, a skinnable clock and one shared Viewer for pictures, video and documents. Shelves get drag-to-open and drag-to-copy, a visible edit mode and sub-shelves. The notification sound is chosen per app. An installation always says whether a restart is needed (almost never).
Format for ever	As well as working in open formats (ODF, PNG, FLAC), Keeper stores a PDF/A and plain text copy of important documents, so they can still be read in 50 years.
First week with Dijana	Seven short tasks in the first week (one a day, including a typing game), after First steps; each can be skipped.

8g. A computer for people

Area	What the OS does
Paper	a greyscale mode without motion, for reading and writing without fatigue; works in both styles
E-ink second screen	Panorama supports an e-ink screen as a second output (slow refresh, partial drawing); Panorama picks the screen face per app, ready for the future hybrid Two-face screen (paper or colour)
Light	automatic brightness and colour warmth by time of day and room light (sensor in the Eye camera)
Focus	with "What are you doing today?": notifications arrive in one batch an hour, and system apps have no hooks (red badges, endless feeds)
Dijana breaks	the 20-20-20 rule (every 20 minutes, look 20 feet (6 m) away for 20 seconds), switchable off, never in the middle of typing
Posture	in the First week the Eye camera helps set the height of the screen and chair, locally and without storing any image
Measurements	usability is measured by an independent lab and the results are public (in addition to the 10-minute test from section 1)

8h. Phone, AI and touch

A phone alongside Neo (optional)

Everything is off until the user pairs a phone. It works locally, with no account and no cloud, using only free software.

Feature	Based on	Android	iPhone
Tether: notifications, clipboard, SMS, remote input, battery, "find my phone"	KDE Connect protocol (GPL)	full	partial (KDE Connect for iOS); notifications over Bluetooth ANCS
Pass to a neighbour: sending files like AirDrop and Quick Share	LocalSend (MIT)	yes	yes
Synced folders (photos from the phone, documents)	Syncthing (MPL)	yes	through third-party apps
Phone in a window: screen and control of an Android phone, the phone camera as a Neo camera	scrcpy (Apache)	yes	no
Calls on Neo (the computer's speaker and microphone)	BlueZ and PipeWire HFP	yes	yes
Calendar and contacts	CalDAV and CardDAV (Diary), DAVx⁵ on Android	yes	yes, built into iOS

```
iPhone stays limited because Apple does not open up messages (iMessage)
or deeper access. Everything above also works in Neo OS Lite.
```

Claude and other AI, by choice

- **Off by default**, with three levels from the first OS document: off, local model only (Ask Neo), and local plus cloud. One switch in Settings and the physical network switch confirm it.
- **Neo MCP bridge:** the Lux ports of every app are already exposed as MCP tools. The user can connect them to the Claude app as a connector wherever their Claude plan allows it (including the free plan, to the extent Anthropic offers that at the time), or through their own API key. Lock asks before any data labelled Personal or Work is sent.
- **Send to Claude:** a Service in the menu that opens the selected text or file in the Claude app, or on the web in Horizon, with nothing sent automatically.
- The same bridge works with other AI assistants that support MCP, and with local models.
- **How much it helps:** separately from where the model runs, the user chooses a help level: **Off**, **Hint** (only points the way), **Explain** (step by step) and **Work with me** (proposes and carries out actions, asking before anything that cannot be undone). In the Kids' room and Neo School the default is Hint.

Touch mode: lessons from Finnish phones

In Neo	What it does	Model
Swipe from the edge	swiping in from the edge closes or puts away a program; there is no "back" or "home" button	Nokia N9 (MeeGo Harmattan)
Three views	Events (batched notifications), Programs and Open, one swipe left or right	Nokia N9
Pull-down menu	pull a list down and a menu opens above it, with a soft tick on each item; works one-handed	Sailfish OS
Live cards	in Open, every program is a card with its state and one or two actions (pause, next track, new document)	Sailfish OS

In Neo	What it does	Model
Peek	a partial swipe shows the Desk or notifications, and letting go returns to the program	Sailfish OS
Ambience	the accent colour and sounds are taken from the background picture, with a contrast check	Sailfish OS
Quick switches	pulling the menu bar down opens Wi-Fi, sound, light, Focus	Symbian Belle
Double-tap to wake	and a clock on the lock screen while idle	Nokia N9

```
Touch mode switches on by itself when a touch screen is connected or the
Neo 500 keyboard is detached; Neo Kit apps get it without extra code.
```

8i. Tiles, the Bar and the Neo menu

Three things people know from other systems, brought together here: **tiles** (small programs on the desktop, like the gadgets in Windows Vista), the **Bar** (a taskbar, transparent if you like) and the **Neo menu** in place of a Start menu. The rule is the same as in the rest of part 8: one system, not three. Tiles, Shelves, the Bar and Helpers belong to one family, so nobody has to learn them separately.

What replaces the Start menu

NeoOS has no single Start menu. Its jobs are shared by four places, each with one job, and all four have been in part 8 from the start:

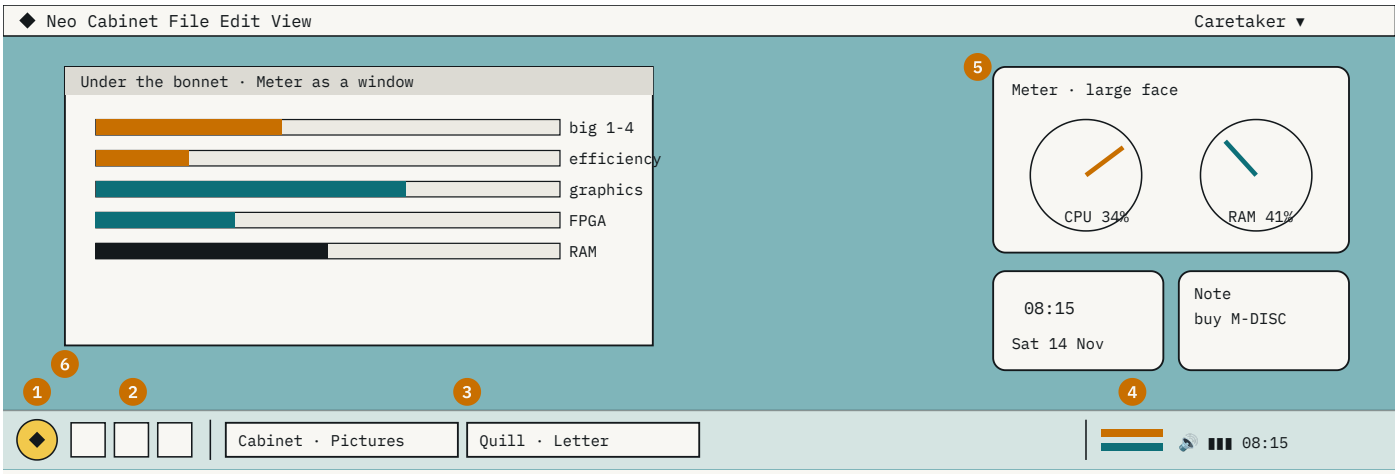
Start menu job	In NeoOS	Where it is
Launch a program or open a document	**Shelves** (and the Bar, which is a kind of shelf)	the ready-made shelf at the bottom, or as many shelves as you like along any edge
Find anything by typing	The **command palette**, through the Find service: programs, files, settings, commands, Lux recipes	the Jump key or Neo+Space, from anywhere
System, recent items, switch off	The **Neo menu** (modelled on the Apple menu): About This Computer, small programs and tiles, recent documents, Nap, Good night, the user's own add-ons	the Neo mark on the left of the menu bar
What is running now	The **Caretaker**: open programs, Hide Others, Under the bonnet	on the right of the menu bar, or in the Bar
Home screen	**"What are you doing today?"** (part 8c): tasks and recent documents as large tiles	at start-up, and a button in the Neo menu

People used to Windows get a **Neo button** at the bottom left of the Bar: it opens the same Neo menu, with the command palette and a search field underneath. It is not a second menu, just the same Neo menu in a different place.

Tiles: the small programs everyone needs

A tile is a small program with three faces, all drawn by the same code. The face depends only on where the user drops it:

Face	Where it lives	Example: Meter
Small (one row, like an icon)	the Bar, a Shelf, the menu bar	two thin bars: processor and memory
Large (square or rectangle)	the Desk, a side shelf, "What are you doing today?"	two dials with needles, as in Vista, plus the busiest chip
Window	double-click any face	the full Under the bonnet view: every core, the chiplets, the FPGA core, RAD:, network, temperature



1 Neo button (the same Neo menu) · 2 pinned programs, as on a Shelf · 3 open windows from the Caretaker · 4 small tile faces and the clock · 5 large faces on the Desk · 6 the same tile as a window

Tiles in 1.0

Tile	What it does	Where the data comes from
Clock	analogue or digital, several time zones, alarm, stopwatch, countdown; the face is chosen like a skin (Silver, Pixel Block, Crystal)	system clock, no network
Calendar	the month at a glance, today's appointments, holidays from the country pack, both Serbian scripts	Diary (CalDAV if chosen)
Meter	processor (big and efficiency cores), memory, graphics, FPGA core, RAD:, network, temperature	Gate statistics, read-only
Notes	sticky notes (the small program from part 8), now also as a tile; searchable through Find	Home:, local
Weather	a forecast for the place the user types in; no location tracking	a public weather service of the user's choice, through Lock; offline it shows the last reading
Battery	Neo 500, headphones, the phone through Tether, the Sun station	native drivers, Tether
Calculator	basic, scientific and programmer (hex, binary); a paper tape with history	local
Also	Mixer and Gramophone controls, RAD:, Sun, Puzzle, Keyboard Map, Scrapbook, Dijana's nook	replicants from programs and small programs

No adverts, no news that arrives by itself and no red badges, under the Focus rule from part 8g. A News tile exists only with sources the user has chosen (part 11a).

Drag and drop

- Any tile can be dragged from the Neo menu (Tiles…), from Treasury, out of a program (a replicant) or from one place to another: the Desk, a Shelf, the Bar, a side shelf.
- While it is being dragged, the tile shows the face it will have where you drop it. Shift makes a copy (two clocks for two time zones).
- Tiles on the Desk snap together edge to edge (like Haiku's Stack and Tile) and are saved in the task's workspace.
- In Easy mode tiles cannot be lost by accident: the Bin brings them back, and the Neo menu has "Put my tiles back as they were".

How they are built

- **Isolated on Fortress.** Every tile is a separate process with its own address space and a memory budget (target: up to 8 MB), like every system program. If one hangs, Fortress restarts it and a grey frame holds its place in the meantime; the other tiles, the Bar and the Desk carry on.
- **Only what it needs.** A tile asks for capabilities in its manifest (Meter: read-only statistics; Weather: network access to one service). Gate grants only that, and Lock asks before the first trip onto the network. The lesson from Vista and Windows 7: their gadgets were retired in 2012 because they had too much access to the system; here a tile cannot do more than an ordinary program, only less.
- **They cost almost nothing.** Live tiles are Panorama layers (part 8d) and refresh at most once a second; when hidden or when the screen is off they do not refresh at all. The Dispatcher keeps them in class 3 (background).
- **One frame in Neo Kit** (`NeoKit.Tile`): a program describes three faces and Neo Kit draws the frame in the Retro, Future Shock or Crystal style. The compositor gets a new extension, `neo-tile`, so Linux programs can offer a tile too.
- **Scriptable.** Every tile has a Lux port ("ADDRESS Clock / ALARM 7:00"), so voice, Ask Neo and the Neo MCP bridge see it like any other program.
- **Anyone can make one.** A tile can be written in Lux or Neo BASIC in about ten lines; Neo School has a lesson called "Make your own clock". Tiles from Treasury are signed, like every package.

The Bar: a taskbar, transparent if you like

The Bar is **a Shelf in a special layout**, not a new system. It has three sections that never mix, so the rule from part 8 still holds ("launching lives on the Shelves and what is running lives in the Caretaker"): on the left the Neo button and pinned programs, in the middle the open windows from the Caretaker, on the right the small tile faces and the clock. It goes along any edge, separately on each Panorama screen, and can hide itself.

Bar look	What it looks like	When
Glass	transparent, blurring what is underneath with a soft sheen, as in Vista; the colour comes from the wallpaper (Ambience), with a contrast check	Future Shock style, Lively or Lavish level
Silver	opaque, bevelled, in the spirit of Mac OS 8 and 9	Retro style, the default
Classic	a grey bevelled bar in the spirit of 1990s home and office computers, the kind RetroBar brings back today	optional; goes with Pixel Block and pixel icons
Crystal	the crystal frame from the third style, if agreed with its author	optional

- **Ray draws the glass** in a compositor layer, so it costs no processor time. On battery, at the Calm level, in the Contrast theme and with "Reduce transparency" (Accessibility) the Bar turns opaque by itself.
- **Neo OS Lite, Neo 100 and Revive**: the Bar is always opaque (Silver or Classic), because blur would cost too much on older graphics chips; tiles still work, with at most six live at once.
- **Touch mode**: the Bar becomes the Open view (the live cards from part 8h), and tiles move to the Events view and the lock screen.

Layouts: the user chooses, the system stays the same

At first start-up (and later in Settings) the user picks a layout. All of them use the same Shelves, tiles and Neo menu, so switching takes one click and loses nothing.

Layout	Top	Bottom	For whom
Neo (default)	the menu bar with the Neo menu and the Caretaker	one Shelf with live tiles	everyone; closest to classic Mac OS
Bar	the menu bar (the global menu stays)	the Bar with the Neo button, windows and clock; the Caretaker moves into it	people coming from Windows
Harbour	the menu bar	a centred Shelf that magnifies icons under the pointer, tiles on a side shelf	people coming from a Mac
Easy	a menu bar with fewer items	a large Shelf with six things and the clock	children, beginners, the Large profile

How it fits with Shelves and Helpers

Shelves grew out of the Amiga dock programs, and Helpers out of Amiga Commodities. Tiles do not replace them; they join them up:

- A **Shelf** is a home: programs, documents, folders, Lux scripts and tiles (small or large face) go on it. The **Bar** is a Shelf with a section for the Caretaker.
- A **tile** is a resident: it has a face and can live in any home, or on its own on the Desk.
- A **Helper** is a tile without a face that works in the background (hotkeys, abbreviations, focus follows mouse). If it wants to be seen it gets a small face (for example a Focus icon in the Bar) and stays the same Helper.
- A **replicant** (from BeOS) is a tile offered by a big program: Gramophone controls or a chart from Abacus are dragged out of the program onto the Desk.

For developers it is one manifest, one Neo Kit frame and one Lux port, whether the thing is a tile, a Helper or a replicant.

```
Estimate: Shelves and Helpers grow from 30 to 50 person-months (the tile
framework and neo-tile, seven tiles, the Bar with four looks, layouts).
The Lite version shares the same code.
```

8j. Neo DOS: the Shell on an equal footing with the Desk

On the Amiga the desktop and the command line were two equal faces of the same system: `LoadWB` started the desktop, `NewCLI` opened a command window and `EndCLI` closed it. NeoOS brings that principle back. Because NeoOS is not Linux (Linux lives only in confined domains, part 3), our own shell has nothing to "mask": **Neo DOS** is the system's native shell, in the spirit of AmigaDOS and TRIPOS, and a Linux shell is an extra for those who want it.

Two faces, always at hand

- The **Shell** is the Neo DOS window. It opens with Neo+Enter, from the Neo menu ("New Shell"), or by typing `>` in the command palette, which runs the command without opening a window.
- **A DOS screen under the Desk:** as with drag-down screens (part 8), pull the Desk down and a full DOS screen is underneath. Neo+Esc lowers it and brings it back.
- **`LoadDesk`** and **`EndDesk`** (`LoadWB` is accepted too, out of habit) start and stop the Desk. When the Desk is stopped, open programs go to sleep through Chronos and come back when it starts again; command windows and services keep running.
- **Starting without the Desk** is a choice at start-up (the D key) and a mode for servers, home servers and repairs. **Neo OS Lite ships with the Neo Desk**, just like the full NeoOS; DOS alone is an option, never the default.

Why the AmigaDOS style

Feature	Neo DOS	Linux shell
Command names	whole words: `Dir`, `List`, `Copy`, `Delete`, `Rename`, `MakeDir`, `Info`, `Protect`, `Search`, `Type`, `Run`, `Execute`, `Avail`, `Status`, `Break`, `Version`, `Which`	abbreviations: `ls`, `cp`, `rm`, `mv`, `chmod`
Arguments	named, from a template: `Copy FROM Home:Pictures TO Archive: ALL`. A `?` after a command shows its template (`FROM/M/A,TO/A,ALL/S,QUIET/S`) and asks for the arguments one by one	letters and dashes whose meaning depends on the program
Disks and places	logical names: `System:`, `Home:`, `RAM:`, `RAD:`, and your own through `Assign Work: Home:Company/2026`; scripts do not depend on where a file physically is	one tree from `/`; mounting needs an administrator
Patterns	`#?`, `?`, `(a|b)`, `~(…)` for "everything except"	globs and regular expressions, different in each tool
Scripts	`Execute` with `If`, `Else`, `Skip`, `Lab`, `Echo`, `Set`, `Ask`; for bigger jobs Lux (in the ARexx spirit) and the `Lux` command	sh, bash
Errors	in plain language, with an NEO-XX-NNN code and a "Why?" button in the Shell	a short message

Additions AmigaDOS did not have: pipes (`Dir Home: | Search "invoice"`, and the `PIPE:` device), history and Tab completion, colour, help for every command (`Help Copy`) and undo: `Delete` sends files to the Bin, and real deletion needs `Delete … FOREVER`.

Command languages

- **Technical names are the same in every language** (`Copy`, `Dir`), as the glossary says, so a script written in Belgrade runs in London and the other way round.
- **Translated names are synonyms** added by a language pack (in Serbian, for example, Latin and Cyrillic words for Copy, List, Delete, Assign and Run), and so are template keywords. `Which` tells you which technical command a synonym stands for.

- Messages and help go through Fluent like the rest of the system (part 1b), so Neo DOS is translated with the same translation kit.

A Linux shell for experienced users

- The **Linux Shell** is a separate shell (bash, zsh or fish, the GNU tools, `ls`, `grep`, pipes) running in the Applications domain (part 3), over the same files. Logical names appear there as paths: `Home:` is `~`, `Work:` is `/neo/Work`.
- It opens with the `Linux` command from Neo DOS, or as the default shell in the Pro profile. From it, the `neo` command runs any Neo DOS command (`neo Assign`).
- Linux programs from Treasury and the Linux box run in it as on any Linux.

Translator: learning in both directions

With `Set Translate on` (or a button in the Shell), every Neo DOS command shows its Linux equivalent before it runs:

```
1.Home:> Copy FROM Pictures/#?.jpg TO Archive: ALL
   ↳ Linux: cp -r ~/Pictures/*.jpg /neo/Archive/
1.Home:> List Documents SINCE yesterday
   ↳ Linux: find ~/Documents -maxdepth 1 -newermt yesterday -ls
```

- **And the other way round:** type `ls -l` in Neo DOS and you get the suggestion "On Neo: `List`. Run it?". In "Linux habits" mode the thirty or so most common Linux commands work directly in Neo DOS.
- **The translation is honest:** it is shown only when the equivalent is exact; otherwise it says "no exact equivalent". Nothing runs on the user's behalf without being shown, and commands that cannot be undone always ask.
- Neo School has a course, "The command line for children", in Neo DOS, with the Translator as the bridge to Linux.

How it fits with the Desk, Shelves and the palette

- The **command palette** takes DOS commands after a `>`; the result appears in a balloon, or opens a Shell if it is long.
- **Shelves and the Bar** accept DOS scripts dragged from the Cabinet; a click runs them (like a tool icon on the Amiga).
- **The Shell tile**: the small face shows the last command and its outcome, the large face is a small terminal on the Desk.
- **Text as a command** (part 8b) works for DOS lines too: a middle click on `Assign Work: Home:Company` in a note runs it.
- Every action in the Cabinet has its own DOS command, and "Record what I do" (part 8f) can produce a DOS script instead of a Lux script.

```
Estimate: Neo DOS (shell, about 45 commands with templates, Assign,
scripts, pipes, help, languages, Translator, starting without the Desk)
35 person-months, on top of the existing 5 for the Shell program. Neo OS
Lite uses the same code.
```

8k. Cabinet Pro: file management in the spirit of Directory Opus

Parts 8 and 8d already give the Cabinet two faces: the **spatial Cabinet** (every folder is a window that remembers its place, like the classic Finder) and **Easy** views for beginners. This part describes a third face for people who work with files a lot: **Cabinet Pro**, modelled on two of the best-loved Amiga programs, Directory Opus 4 and Directory Opus 5 "Magellan". Both are open source today (Opus 4 under the GPL since 2000, Magellan since 2012), so we can take the ideas freely, while the name and the look are our own.

What we take from Opus 4 (and Norton Commander): the Duo layout

Duo is the retro mode of Cabinet Pro: two panels side by side, with a button bank between or below them. One panel is the source and the other the destination, as in Norton Commander, and the buttons work as in Opus 4.

Feature	Model	In Duo
Two panels, source and destination	Norton Commander, Opus 4	Tab switches the active panel; each panel has a path field (with logical names: `Home:`, `Work:`) and buttons for parent, root and disks.
Button bank	Opus 4	Ready-made buttons for the basic jobs: Copy, Move, Rename, Delete, New folder, Info, View, Play, Compare, Find, Size, Protect, Comment, Date, Crunch, Archive, Unpack, Hex view. Several banks (page 1, 2, 3) switch with one click.
Left and right click on a button	Opus 4	Every button has two actions: for example a left click for "Copy" and a right click for "Copy as..."; a middle click shows what the button runs.
Function keys	Norton Commander	F3 View, F4 Edit, F5 Copy, F6 Move, F7 New folder, F8 Delete (to the Bin), F10 Quit; a strip with their names along the bottom.
A command line under the panels	Norton Commander	Neo DOS (part 8j) in the same window; `{selected}` and `{other}` are replaced by the selected files and the other panel's path.
Folder buffer	Opus 4 "buffer list"	Recently opened folders stay loaded, so going back is instant; the list opens with a click on the path.
"Will it fit?"	Opus 4 "CheckFit"	Before copying, it shows whether the selection fits on the destination (USB stick, cartridge, M-DISC) and how much is left.
Find and Size	Opus 4 "Hunt", "GetSizes"	Finding a file across a whole disk and the real size of a folder, in the background.
Duo without the Desk	Norton Commander on DOS	The same Duo also runs as a text program on the DOS screen (part 8j), for servers, repairs and Lite without the Desk.

What we take from Magellan: the Cabinet as a whole working environment

Magellan replaced the whole Amiga desktop, not just the file manager. In NeoOS the Cabinet is already part of the Desk, so from Magellan we take what makes that Desk more powerful:

Magellan feature	In NeoOS
A lister that changes face: the same window as icons or as a list with columns	Every Cabinet window changes view with one key (icons, list, columns, gallery), and two windows join into a Duo with one move.

Magellan feature	In NeoOS
Floating button banks	These are our **Shelves** (part 8i): a button bank from Duo can be torn off and stand on the Desk or along an edge as a Shelf.
Filetypes with triggers: what happens on double-click, drag, right-click	**Datatypes** (part 4) get actions per event: a double-click opens, dragging onto a "Print" Shelf prints, a right-click gives a menu of actions for that type. Users add their own.
User menus in the menu bar	A "Mine" menu in the menu bar and add-ons in the Neo menu (the `Settings:Neo menu` folder), with Neo DOS commands and Lux scripts.
Scripts on events: disk inserted, start-up, file dropped	**Triggers**: "when a USB stick with photos is inserted, offer to import them into Album", "when a PDF is dropped into the Invoices folder, crunch it and label it Personal". Written as Lux or Neo DOS scripts, or through "Record what I do".
FTP as a lister	Already in the Cabinet: FTP, SFTP, network disks, the phone (through Tether) and archives as folders.
Groups and icons left out on the desktop	Piles on the Desk and Case (part 8d); any file can be "left out" on the Desk as a link.
Themes and hotkeys	The Retro, Future Shock and First Desk styles, and a hotkey for every button and menu item.

Everything is programmable

- **A button is a command.** Every button, menu item and filetype action is one line of Neo DOS or a Lux script, so whatever a button does also works from the Shell, by voice and through the Neo MCP bridge.
- **Bank editor:** drag a command from the Shell or a script from the Cabinet onto a bank and it becomes a button; name, icon, colour and hotkey are chosen on one properties sheet.
- **Sharing:** banks, filetypes and triggers export as one small file and are shared through Treasury and Co-op clubs, signed like every package.
- **Safe:** a button that deletes or sends something over the network shows what it will do and asks first (Lock for the network, the Bin for deletion).

How it fits

- **Lisa's principles stay:** in Easy mode there are no buttons or panels, only documents, pads and views (parts 8b and 8d). Cabinet Pro is for the Pro profile, and anyone can switch it on.
- **Object, then action** (part 8b): first choose the file, then the button; the same order everywhere.
- **Neo DOS** is the language of the buttons, and the Translator (part 8j) shows the Linux equivalent here too.
- **Crunching** (part 7d) and archives are ready-made buttons; the Disk tile leads straight into Duo.
- **The full NeoOS and Lite** have the same Cabinet Pro; Lite draws Duo in the Retro style, and text-mode Duo works without the Desk.

```
Estimate: Duo (window and text mode), button banks and editor, filetype
actions, triggers, user menus, sharing: 30 person-months, on top of the
existing 45 for files in Easy mode and Cabinet Pro modules.
```

9. Compatibility and retro

Runs almost everything. A double-click picks the room by itself: Bridge (Wine and Proton), full Windows on Satellite, the Linux domain, the Android room (with microG, no Google account), Chameleon and Arcade, or the web. The **"Does it work?"** database in Treasury records users' experience with each program. What does not run: games with kernel-level anti-cheat, macOS and iOS programs, and Android apps that require Play Integrity.

Layer	What it runs	How	1.0
Arcade	retro games and systems	An FPGA core in Chameleon if one exists, otherwise an emulator (libretro, MAME), otherwise a box. The user chooses a game, not an emulator.	yes
Bridge	x86 Linux and Windows programs and games on ARM	FEX and Box64 for translation, Wine and Proton, DXVK and VKD3D-Proton	yes
Satellite	a second processor on a card: x86 (Windows, DOS), RISC-V or Power (AmigaOS 4 and MorphOS with the buyer's own licence)	CXL link, picture as a Panorama layer, an agent for clipboard and files; Autoconfig recognises the card type	x86 yes, others 1.1
PowerPC room	AmigaOS 4 and MorphOS without a Power card	PowerPC emulation on ARM (QEMU), with the buyer's own licence for the system	1.1
Link	Windows windows as Neo windows	an agent inside Windows, windows through the compositor, shared clipboard	yes
Linux box	Flatpak programs	Wayland, they get Neo frames and the menu bar	yes
Android room	Android programs	Waydroid with microG	1.1
ReactOS box	Windows without a licence, experimental	whole-system emulation	2.0

Limits to state up front: games with kernel anti-cheat run only on Satellite with real Windows; Netflix and similar services at full resolution depend on a Widevine licence; Adobe and AutoCAD run through Satellite or Bridge, with varying success.

10. Applications in 1.0

Priority: P0 must be in 1.0, P1 goes into 1.0 if it passes the beta gate, P2 goes into 1.1 or later. P-M is the estimate in person-months. Every app uses Neo Kit, has an Easy/Pro switch and a Lux port, and works without an account.

Application	What it does	Engine	P-M	Priority
Neo School and Classroom	Offline lessons, assignments and lessons in class	Kolibri, our own	42	P0
Playroom	Educational games for children	GCompris	6	P0
Neo Blocks	Programming with blocks	Snap!	20	P0
Lab	Science simulations	PhET	8	P0
Sky	Planetarium	Stellarium	6	P1
Recall	Flashcards	Anki format	10	P1
News	Sources the user picks, no algorithm	Newsflash core, RSS/Atom	10	P0
Cabinet	Files, two panes, archives and FTP as folders	Haiku Tracker, libarchive, GVfs	14	P0
Horizon	Web browser	WPE WebKit, parts of Epiphany	32	P0
Pigeon	Mail with delayed sending	Geary core, JMAP	14	P0
Quill	Word processing	LibreOffice core, Neo Kit look	30	P0
Abacus	Spreadsheets	LibreOffice core	30	P0
Canvas	Drawing and layout	LibreOffice Draw core	16	P1
Stage	Presentations	LibreOffice Impress core	16	P1
Archive	Database for home and small business	SQLite	24	P1
Leaf	PDF, EPUB, comics, DjVu	Poppler, Foliate	8	P0
Album	Photos, faces and places, locally	darktable, libvips	12	P0
Gramophone	Music, radio, podcasts	GStreamer	8	P0
Cinema	Video, subtitles and local translation	mpv	6	P0
Wave	Audio editor	Audacity engine, LV2	12	P1
Diary	Notes, calendar, contacts	Evolution Data Server, CalDAV	12	P0
Atlas	Offline maps	Organic Maps	8	P1
Talk	Messages and video calls	Element/Matrix SDK, Jitsi	10	P1
Keyring	Passwords in the Safe chip	KeePassXC format	6	P0
Scanner	Scanning and OCR, Latin and Cyrillic	Tesseract	6	P1
Interpreter	Offline translation	Bergamot, Neuron	8	P1
Tether	Phone with Neo: notifications, files, folders, phone in a window, calls	KDE Connect, LocalSend, Syncthing, scrcpy, BlueZ	35	P0
Arcade	Games and cores: FPGA, emulator or box	RetroArch, libretro, MAME	12	P0
Shell	Terminal that explains commands	fish shell ideas, VTE	5	P0
Neo BASIC	A language for first programs	SmallBASIC core	12	P0

Application	What it does	Engine	P-M	Priority
Neo Cards	Card-based apps in the spirit of HyperCard	Neo Kit	16	P2
Neo Studio	Visual app building	Neo Kit	40	P2
Cut	Video editing on proxy footage	Kdenlive and MLT	20	P2
Neo Workshop	Pro pack: Sculptor, Brush, Vector, Type, Darkroom, Cut Pro, Studio, Director	Blender, Krita, GIMP, Inkscape, Scribus, darktable, Ardour	60	P2
Broadcast	Streaming with Panorama scenes	OBS Studio	8	P2
Neo Support	The team's channel in the OS: bug reports with consent, adaptation status, expert help	our own	15	P0
Keeper	Backups	Btrfs snapshots, restic	8	P0
Teletext and BBS	Teletext-style news, BBS client	our own	7	P1

Neo Workshop is a Pro pack for €99 to €149, after 1.0: Sculptor (Blender), Brush (Krita and GIMP), Vector (Inkscape), Type (Scribus), Darkroom (darktable), Cut Pro, Studio (Ardour) and Director (ArtCraft, licence under review). Half the income goes to the original projects. The work is mostly adapting them to the Neo Kit look, Datatypes and Lux ports.

```
612 P-M in total for everything listed, of which 468 P-M for P0 and P1.
Quartet (Quill, Abacus, Canvas, Stage) is built on the LibreOffice core
with a new Neo Kit look, which saves about three quarters of the work
compared with writing from scratch.
```

11a. Learning comes before games

Neo is above all a computer for learning, like the Galaksija and the BBC Micro. Learning takes priority over games in the plan, in the box and in "What are you doing today?".

In Neo	What it does	Based on	1.0
Neo School	lessons by school year with no internet, from the Knowledge cartridge	Kolibri (MIT)	yes
Classroom	the teacher hands out and collects work, sees progress and locks screens during a lesson; also works without internet through the Neighbourhood	our own	yes (previously 1.1)
Playroom	over 150 educational games for children aged 2 to 10	GCompris (GPL)	yes
Neo Blocks	programming by snapping blocks together, with a path on to Neo BASIC and Lux	Snap! (AGPL)	yes
Neo BASIC and **Shell**	first real programs and working with the system	SmallBASIC	yes
Lab	simulations in physics, chemistry and mathematics	PhET simulations (GPL and CC BY)	yes
Sky	planetarium	Stellarium (GPL)	yes
Recall	flashcards with spaced repetition at the right moment	Anki format (AGPL)	yes
Library and **Ask Neo**	Wikipedia, books and answers with sources, without internet	Kiwix	yes
Pin header and **FPGA area**	electronics and digital logic for older pupils	section 7b	yes

News you choose

News is a reader for sources the user picks (RSS and Atom), in time order. There is no algorithm pushing content, no "recommended", no adverts and no sources set up in advance by anyone. The first time, News shows an empty list and offers to add sources by hand or from an open catalogue. There is also a Teletext-style view with page numbers.

11. Games and FPGA cores in 1.0

Launch games have two jobs: to entertain and to show off the hardware (Newton, Copernicus, T-Buffer, Chronos rewind). Every game works with a DE-9 joystick and the Neo remote.

Game	Description	P-M	Priority
Crystal	Original puzzle game with hexagonal blocks	8	P0
Patience and Minesweeper	Classics on the Aisleriot base, with the Neo look	3	P0
Grandmaster	Chess with lessons and analysis (Stockfish)	12	P0
Pinball Cosmos	Three tables, shows off Newton physics and the T-Buffer	30	P0
Midnight Poker	Poker for the 18+ profile, no real money, network and AI	10	P1
Forest Bane	2D platformer, about 30 levels, Dijana as guide	40	P1
Swarms	Arcade shooter with Copernicus effects; if a licence for a known game is agreed, this becomes its port under our name	14	P1
Depths	Metroidvania: a licensed port of a known title, sold under our name	12	P2
Ras (fallback)	Our own metroidvania if the licence falls through	90	after 1.0
Start-up and demo	Demoscene for the first power-on and the screen saver	6	P0

FPGA cores for Arcade

FPGA core	Based on	P-M
Amiga OCS/ECS/AGA	Minimig-AGA + TG68	10
Commodore 64	with SID	4
ZX Spectrum and Z80 (SymbOS)	—	5
NES, SNES, Mega Drive	—	10
1980s arcade machines	—	8
Atari ST	—	5
1996 3D accelerator	open FPGA core (SpinalVoodoo), cycle-accurate for old Glide games	8

The cores are adapted from existing open projects (Minimig-AGA and TG68 for the Amiga, MiSTer-class cores for the rest), respecting their licences. Commercial ROMs are not shipped.

> Licensed ports (Swarms and Depths, if agreed) depend on agreements with the rights holders. The plan does not count on them as a condition for 1.0; the fallback is Ras, our own game, which would come after 1.0.

12. Developer tools

- **Neo SDK**: Neo Kit for C++ and Rust, Lux, Vulkan and Neo 2D, the Copernicus, Newton and Neuron libraries, and the neopak packaging tool.
- **Neo simulator**: the whole of NeoOS in QEMU on an ordinary PC or Mac, with software models of Copernicus and Chronos. Developers start before the hardware exists.
- **Neo BASIC** in the box (on the SmallBASIC core), with graphics, sound and joystick in one line.
- **Treasury for developers**: publishing from a git repository, reproducible builds, no commission on free programs.
- **Documentation** in English and Serbian, with examples that run from the Shell.
- **Language SDK** (section 1b): everything a country or community needs to add its own language: strings, fonts, keyboard layouts, voice, everyday names and a style guide.

12b. The open source we use

The rule: anything that exists as a mature open project under the GPL or a compatible licence is taken as the foundation, and the Neo team writes only what is genuinely new (our chips, the look, Lux, Tasks, Chronos, Lock). Changes go back to the original projects, so the community maintains them together with us. Outside projects stay inside the system; the user sees only Neo names.

Part of NeoOS	Based on	Licence	Saving
Fortress, Exec-N, Gate	Genode OS framework on the seL4 microkernel	AGPLv3 / GPLv2 (seL4)	about 90 P-M
Native drivers for bought-in chips	Linux driver code ported into isolated native components (the Genode DDE method)	GPLv2	about 95 P-M
Network stack	Linux TCP/IP as a component, lwIP	GPLv2, BSD	15 P-M
File system and mirror	Btrfs, EROFS	GPLv2	15 P-M
Vulkan, OpenGL, OpenCL	Mesa (NIR compiler, common Vulkan runtime, Zink, Rusticl)	MIT	60 P-M
Compositor	Smithay (Rust Wayland library)	MIT	25 P-M
Neo Kit	Haiku Interface Kit as a starting point, ideas from Zune	MIT, APL	30 P-M
Sound server	PipeWire	MIT	15 P-M
Voice and Ask Neo	whisper.cpp, Piper, llama.cpp, Kiwix	MIT, GPLv3	20 P-M
Bridge and boxes	FEX, Box64, Wine, Proton, DXVK, VKD3D-Proton	MIT, LGPL, zlib	35 P-M
Accessibility	ideas from Orca, eSpeak NG, AT-SPI	GPL, LGPL	15 P-M
Find and Treasury	Xapian, OSTree, Reproducible Builds tools	GPL, LGPL	30 P-M
SDK and simulator	QEMU, LLVM, Rust	GPLv2, Apache	15 P-M
Tests	Khronos CTS, LAVA, ideas from KernelCI	Apache, GPL	15 P-M
Applications	Haiku Tracker, Epiphany, Geary, Organic Maps, Element, Kdenlive, OBS, RetroArch, SmallBASIC, Aisleriot	GPL and related	about 60 P-M
FPGA cores	MiSTer cores, Minimig-AGA, TG68, SpinalVoodoo	GPL and related	already counted

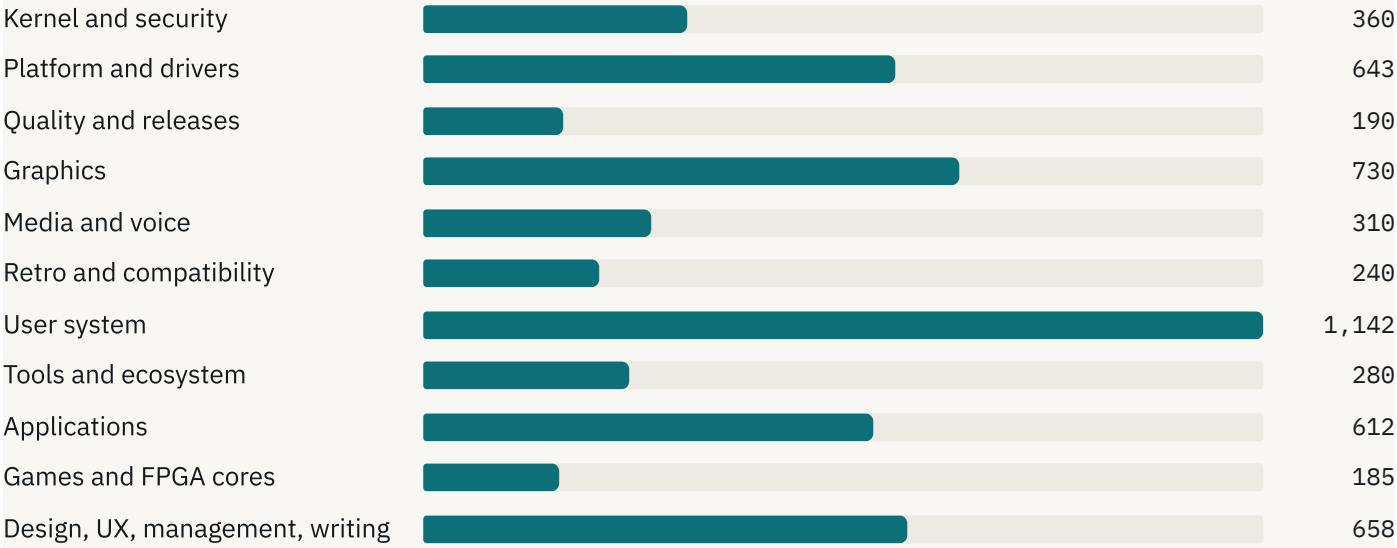
The licences fit together: NeoOS is under the GPLv3 or later, which accepts MIT, BSD, Apache 2.0, LGPL and GPLv2-or-later code. Parts available only under the GPLv2 (Linux driver code) and the AGPLv3 (Genode) live as separate components under their own licence. A legal review of the licences is part of phase F0.

Genode also changes Fortress's fallback plan: Genode runs on other kernels too (NOVA, its own base-hw), so if seL4 fails the month-12 gate, only the kernel underneath changes, and the drivers and services stay the same.

PART B · DEVELOPMENT ESTIMATE

13. Size estimate

Method: every component is estimated in person-months (P-M), already including the open-source savings from section 12b, by analogy with known projects (the Asahi and Turnip Mesa drivers, seL4 systems, Haiku, SteamOS, LibreOffice Online), and 14% is then added for design, UX, writing, translation and management. The estimate covers software only; hardware, cases and manufacturing have their own plans.



5,350 P-M in total: engineering 4,692, design and management 658. The largest single item is the Vulkan driver for the VSA-6000, about 320 P-M.

Platform by component

Area	Component	P-M
Kernel and security	Boot loader, Neo ROM, A/B image and Renewal	60
Kernel and security	Fortress: microkernel, VMM, framework for Rust services	110
Kernel and security	Exec-N (messages, ports, shared memory)	35
Kernel and security	Swarm (small jobs, work stealing, chip offload, learning), 4 Dispatcher classes, Full throttle	75
Kernel and security	Lock, Safe, capability sandbox, MTE	80
Platform and drivers	Native drivers: PCIe, AGP-X, CXL, USB4/xHCI and USB classes, NVMe, ROM, HDMI input	120
Platform and drivers	Native drivers: Wi-Fi 7, Bluetooth 5.4, 5G modem, Ethernet	85
Platform and drivers	Native network stack (Rust) with Lock	25
Platform and drivers	Linux domains (Peripherals, Storage, Applications) and paravirtual channels	45

Area	Component	P-M
Platform and drivers	Courier, cartridge, NVMe, power, solar mode	60
Platform and drivers	Storage: mirror, index, assigns, file versions, RAD: disk	48
Platform and drivers	Neo card: recognition, one image for all cards, Autoconfig, Under the bonnet	50
Platform and drivers	Pin header and user FPGA area from Lux	20
Quality and releases	Portability to RISC-V: platform layer, CI and tests in QEMU	35
Platform and drivers	Dispatcher, chiplet power gating, memory without swap, RAD: fast layer	55
Platform and drivers	Chronos service (snapshot, rewind, rollback)	50
Platform and drivers	Crunching: crunched programs in the loader, dictionaries, compression in Storage, Crunch and Decrunch, PAR2, the Disk tile (part 7d)	15
Graphics	VSA-6000 Vulkan 1.4 driver (Mesa), T-Buffer	270
Graphics	Zink (OpenGL/ES), Glide layer, WebGPU support	20
Graphics	Native kernel drivers for VSA-6000, Ray, Panorama (Rust)	90
Graphics	Panorama and Ray: display, layers, blitter, CRT mode	110
Graphics	Console ideas: object list, transparency, microcode, Pocket, smart cartridge, Arcade Cabinet	65
Graphics	Compositor: Wayland + neo-menubar/layer/snapshot	55
Graphics	Newton, Copernicus and Neuron: drivers and SDK	120
Media and voice	Orpheus: driver, sound server, Paula mode, HRTF	45
Media and voice	Video (AV1), Eye camera with native ISP driver	90
Media and voice	Voice (dictation and commands in Serbian and English), Ask Neo, Library	70
Retro and compatibility	Chameleon (FPGA service, core signing)	40
Retro and compatibility	Bridge: FEX/Box64, Wine, Proton, DXVK	60
Retro and compatibility	Link and Boxes (Linux, Android)	45
Retro and compatibility	Satellite service for x86, RISC-V and Power cards, PowerPC room	70
User system	Neo Kit (controls, layout, Easy/Pro, themes)	120
User system	Desktop, menu bar, palette, notifications, Settings	90
User system	Lux language, ports and MCP bridge	45
User system	Tasks, Pads, Jump, universal commands, Newton-style store	55
User system	Shelves, Helpers, tiles, the Bar and layouts (part 8i)	50

Area	Component	P-M
User system	Neo DOS: shell, template commands, Assign, scripts, pipes, languages, Translator, running without the Desk (part 8j)	35
User system	Cabinet Pro: Duo (window and text mode), button banks and editor, filetype actions, triggers, user menus (part 8k)	30
User system	Lisa principles (the Desk coming back, documents following their disk, window and icon) and the First Desk theme with its sound set (part 8b)	12
User system	Device and user profiles (Pocket, Large, Shared computer)	50
User system	Languages: Fluent messages, Serbian Latin/Cyrillic, en-GB/en-US, Weblate translation kit	25
User system	Touch mode (edges, three views, pull-down menu, live cards, peek, ambience)	40
User system	Neo MCP bridge for Claude and other AI, Send to Claude, three levels and Lock	15
User system	A computer for people: Paper, e-ink screen, light, Focus, breaks, posture	45
User system	Even easier: input to suit the job, Guest USB, Call your own, Record what I do, Progress, shared parts and Viewer, Format for ever, First week	160
User system	Two styles: Retro (Pixel Block, pixel icons) and Future Shock, per-app exceptions	40
Platform and drivers	From the Unix world: preview, Recipes, Carry on, One mouse, Move house, Disk mode, mDNS, IPP, portals, Neo in your pocket	70
User system	Dijana resident and guide, system face, Dijana's day, happenings on the Desk	55
User system	Everyday language (Easy/Pro names), What are you doing today?, First steps, simplicity rules	30
User system	Residents (framework, SDK tool, purchase in Treasury) and personal touches	40
User system	Datatypes (1,000+ formats, sandboxed decoders), Undo, Scrapbook, Find	70
User system	Beauty without the weight: glass, live tiles, window physics, Flip through, levels, Under the bonnet, Shop window	45
User system	Files in Easy mode (links, views, piles, Case, Bag) and Cabinet Pro modules	45
Media and voice	System sound: spatial event sound, ducking, captions, sound themes	25
Tools and ecosystem	Co-op: borrowed power, shared build cache, clubs, Board, Neighbourhood mail (LoRa)	70
Retro and compatibility	Runs almost everything: automatic room choice, Does it work? database	25
User system	Accessibility and localisation (Latin, Cyrillic, English)	45
Tools and ecosystem	SDK, simulator, documentation	55
Tools and ecosystem	Road to a standard: translation kit, factory tools, Neo test kit, national layer	50
Tools and ecosystem	Treasury: neopak, reproducible builds, store	60
Media and voice	Neo Sound: speaker and headphone profiles, crossover, loudness and protection, listening modes and pure output, LE Audio and Auracast, hearing diary and check	35
Media and voice	Sound library: public domain and CC0, rights data per country	20
Media and voice	Audiobooks: LibriVox, Read to me in a local voice, family library, DAISY	25

Area	Component	P-M
Tools and ecosystem	Country packs: framework for local libraries, voice, music and holidays	15
Tools and ecosystem	Protected localisation delivery: two signatures, encryption, cartridge, LAN and network, log	20
Tools and ecosystem	Partner tools for adaptation (Weblate workspace, in-context review, pilot schools)	10
Quality and releases	CI, hardware lab, tests, Vulkan CTS	115
Quality and releases	External security audits and fixes	40

14. Teams

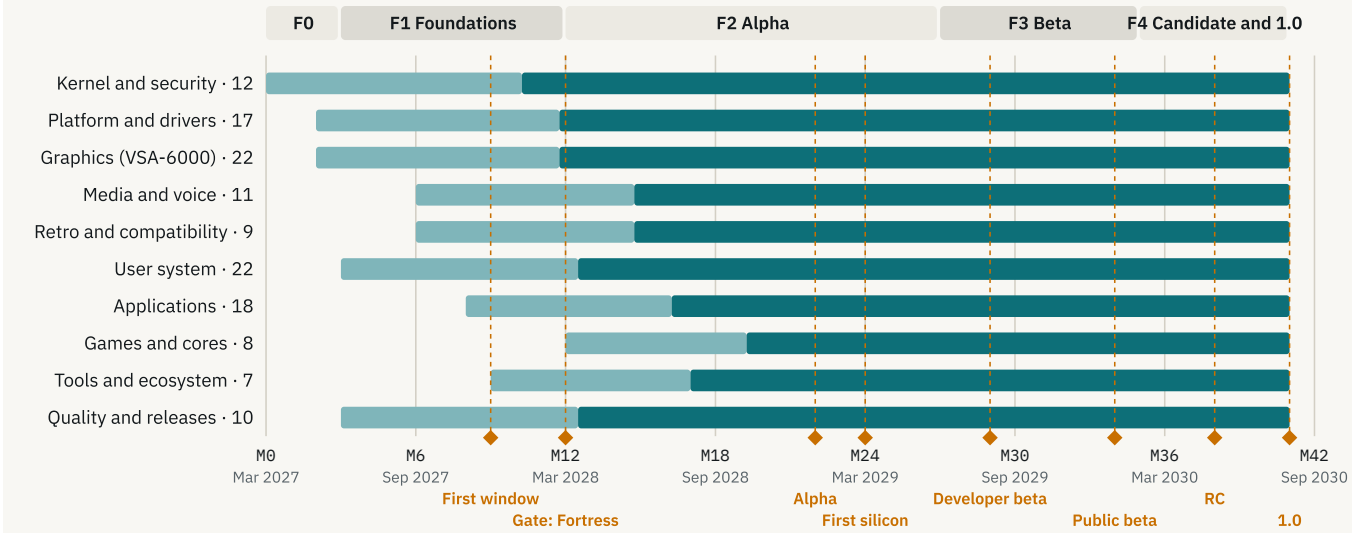
Ten engineering teams and two small groups for design and management. The number is the largest headcount of the team, reached around month 12; teams grow during the first months.

Team	People	Responsible for	Rare skills
Kernel and security	12	Fortress, Gate, Exec-N, Lock, Safe, Renewal, boot loader	seL4, formal verification, Rust
Platform and drivers	17	native drivers for buses, storage, network and wireless, Courier, cartridge, Dispatcher, Chronos, power, Linux domains	Rust drivers, Wi-Fi and 5G, PCIe and CXL
Graphics	22	native kernel drivers and VSA-6000 Vulkan, Zink, Glide, Panorama, Ray, compositor, Newton and Copernicus APIs	GPU drivers, shader compilers
Media and voice	11	Orpheus, video, Eye, voice, Ask Neo, Neuron, Neo Sound, sound library, audiobooks	ISP tuning, speech models for Serbian and English
Retro and compatibility	9	Chameleon, FPGA cores, Bridge, Link, boxes, Satellite	FPGA (Verilog/VHDL), Wine
User system	22	Neo Kit, desktop, Tasks and Pads, Settings, Lux, Datatypes, accessibility, language SDK	toolkit design, screen readers
Applications	18	two groups: everyday apps and learning; Quartet, Horizon and Pigeon	LibreOffice and WebKit cores
Games	8	launch games, cores for Arcade, ports	games on custom hardware
Tools and ecosystem	7	SDK, simulator, Treasury, documentation, developer relations, country packs and adaptation partners	—
Quality and releases	10	CI, hardware lab, automated tests, releases, security audits	—
Design and UX	8	Neo Silver, icons, the Neo System font, sounds, text and messages, Dijana's animation and screen saver, learnability tests	pixel and type design
Management	4	product, plan, licences and legal matters	—
Total	**148**		

The hardest people to hire are for GPU drivers, wireless drivers, seL4 and FPGA. That is why the first 6 months of the plan include working with the Mesa, seL4 and MiSTer communities, and contracts with firms already active in those fields.

15. Phases, gates and timeline

M0 is the start of work, assumed here to be March 2027. Native drivers for all hardware add about 3 months, and the open source from section 12b gives back about 2; it cannot be shortened further, because the date is held by the arrival of silicon. The lighter part of each bar is the time when the team is growing. Orange marks are gates: points where we check whether the plan still holds.



team growing full strength gate

Phases

Phase	Months	What is done	Exit (gate)
F0 Preparation	M0 to M3 Mar to Jun 2027	Hiring the core of each team, development platform, simulator, CI, licence decisions	Fortress runs a Linux domain on the development board
F1 Foundations	M3 to M12 to Mar 2028	Exec-N, Gate, domains, Neo Kit alpha, compositor on an off-the-shelf GPU, first programs, VSA-6000 driver on a model	M9 **First window**: Cabinet in the Neo Silver look. M12 **Fortress gate**: the targets from section 3, or a move to another kernel under Genode
F2 Alpha	M12 to M27 to Jun 2029	All services, P0 apps, Bridge, Arcade with cores, VSA-6000 on FPGA emulation, voice	M22 **Alpha**: the team works on NeoOS every day. M24 first silicon
F3 Beta	M27 to M35 to Feb 2030	Bring-up on real silicon, all native drivers on all three models, speed, Vulkan CTS, P1 apps, games, Treasury opens	M29 **Developer beta** (SDK, Treasury). M34 **Public beta**
F4 Candidate and 1.0	M35 to M41 to Aug 2030	Fixes only, security audit, accessibility, translation, factory image for ROM	M38 **RC**. M41 **NeoOS 1.0**, together with the computer's launch

Conditions for 1.0

- All targets from section 1 met on all three models.

- Zero open P0 bugs, fewer than 20 P1 bugs, all with a known workaround.
- 72 hours of load testing without a domain crash, 500 update-and-rollback cycles.
- An external security audit of Fortress, Gate, Safe and Treasury with no open critical findings.
- A public beta of at least 3 months, with the most common reports fixed.
- The English (UK and US) and Serbian (Latin and Cyrillic) versions complete and checked by native speakers, with no untranslated strings.

16. Scenarios and budget

Scenario	1.0 after	Date	Assumptions
Optimistic	34 months	Jan 2030	silicon arrives on time with no respin, Fortress passes its gate, hiring to plan
Realistic	41 months	Aug 2030	the plan in section 15: one minor silicon fix, somewhat slower hiring for GPU work
Pessimistic	52 months	Jul 2031	a new silicon revision needed (about 9 months), a move to another kernel, a delay with the wireless chip driver

Open question: which card 1.0 ships on

With the swappable Neo card (section 7b), the first card for customers carries an off-the-shelf ARM SoC and an FPGA (Phase 1), and the Neo package with graphics chiplets comes later. The native driver rule applies to both cards. Two paths:

Path	1.0	What it means
1.0 on Phase 1 (proposed)	about M30, Sep 2029	native drivers for the chosen SoC and its graphics (about 100 P-M more; the Vulkan part exists in Mesa); support for the Neo package card arrives in 1.5, around M41. Customers get the computer a year earlier and later swap only the card.
1.0 on the Neo package (the plan in section 15)	M41, Aug 2030	as described in this specification; the Phase 1 card is used only for development

Outline software budget

Item	Estimate	Basis
People	€34 to 40 million	5,350 P-M × €6,500 to 7,500 total cost per P-M (a mix of Serbia and the EU)
Hardware lab, development boards, FPGA emulation	€2 to 3 million	about 400 development units, pre-silicon emulator
Licences (codecs, aptX/LDAC, game ports, fonts)	€1 to 2 million	depends on agreements
Independent usability lab	€0.2 to 0.3 million	public measurements before the beta and before 1.0
Security audits and certification	€0.5 to 1 million	two external audits
Total	**about €38 to 46 million**	**excluding hardware and manufacturing**

The amounts are not verified quotes; they show the order of magnitude.

17. Risks

Risk	Likelihood	Impact	What we do
The VSA-6000 Vulkan driver is late (the biggest risk)	high	high	The driver team sits with the chip team from day one, the driver is written on FPGA emulation before silicon, and Mesa is the base for the Vulkan part. The 1.0 date moves with the silicon, because launching without the VSA-6000 is not an option.
Neo package silicon is late or needs a fix	medium	high	The OS does not depend on silicon until M24; simulator and development platform; drivers are written on FPGA models of the chips.
Fortress with Linux domains is too slow or too complex	medium	medium	A gate at M12 with clear targets; Genode on another kernel, then pKVM, as fallbacks with the same domain layout.
Native drivers for bought-in chips (Wi-Fi 7, 5G, Bluetooth)	medium	medium	A chip is chosen only with public documentation or an access agreement; the chip maker as a partner; a fallback chip of the same class chosen in advance.
Shortage of specialists (GPU, seL4, FPGA)	high	medium	Early hiring, working with the communities, contracts with specialist firms, remote work.
Too many apps for one release	medium	medium	Strict P0, P1, P2 priorities; a P1 app that does not pass the beta goes to 1.1 without moving the date.
x86 translation not fast enough	medium	medium	Hardware TSO support in the ARM chip as a condition for choosing the chip; Satellite for demanding programs.
Licences (game ports, Widevine, HEVC, aptX)	medium	medium	Our own games as the base, AV1 as the main codec, licences as an extra rather than a condition.
Voice quality in Serbian	medium	low	Fine-tuning on open Serbian corpora, working with universities; the keyboard always works. English voice uses existing open models for UK and US accents.
Supply-chain attack on Treasury	low	high	Reproducible builds, signatures in Safe, separate keys for building and publishing, a public release log.
Legal questions about the look (similarity to Mac OS)	low	medium	Everything drawn in-house, our own names (no third-party or licensed name anywhere in the system), a lawyer's review before the beta.
Midnight Poker and gambling rules	low	low	No real money and no buying chips, 18+ profile only.

18. Road to a standard

Neo is not just one product but a proposed standard: a country that adopts it in schools and administration gets a free OS, a licence for local assembly and a translation into its language, and commercial manufacturers pay for the "Neo" name and a certificate. The models are the Pravec, the Galaksija and the BBC Micro, and today China's move to its own OS and chips. The whole model is in the overview "Neo family of computers"; here is what it requires from NeoOS.

Requirement of the standard	What NeoOS must have	1.0
Free and open OS	All of NeoOS, Neo Kit, Lux and Quartet under the GPL and related free licences, with a public repository and reproducible builds. Anyone can check that the image in ROM was built from the published code.	yes
Translation into the country's language	The language SDK from section 1b: all strings in one place, a tool for translators, a check that text fits in buttons, support for right-to-left scripts, Cyrillic, Greek, Georgian and Indic scripts. Voice and Interpreter get models for the new language through Treasury.	tool yes; first languages: English (UK, US) and Serbian
Local manufacturing	Factory tools: writing the Neo ROM, testing every unit on the line, pairing the Safe chip. The same native drivers, because a licensed manufacturer builds the same board from certified parts.	yes
Certificate for manufacturers	The Neo test kit: an automatic check that a manufacturer's computer does everything the original does (drivers, the targets from section 1, Arcade, Bridge). Only those who pass may use the "Neo" name.	yes
A national update channel	Renewal accepts an extra signed layer (school software, national certificates, translation) on top of the system image. The keys for that layer are held by the country, but they do not unlock users' data: users' Safe keys belong only to the users, and that does not change for anyone. The same privacy promise holds in every country, with no hidden tracking and no exceptions for any authority.	yes
Schools and the public sector	Classroom and Neo School (section 11a): the teacher hands out and collects work, the Kids' room, profiles without internet, the Library and the Knowledge cartridge as an offline textbook.	yes
Countries under sanctions	They receive only publicly released code; Treasury services and signed images follow legal advice.	rules in 1.0

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Extra work: about 50 P-M (language SDK, factory tools, Neo test kit, the
national layer in Renewal), counted under Tools and ecosystem. Classroom
has moved into 1.0 (section 11a). The 1.0 date does not change.
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18b. The education and licensing offer

This is what an offer from Neo to a ministry, a school authority or a manufacturer would look like. It is important to be honest: NeoOS is free software under the GPL, so anyone may translate and change it without paying. What is paid for is not the code but **the "Neo" name, the certificate, signing in the Renewal channel, support and the Neo team's work**.

Three packages

Package	For whom	What they get	Indicative price
Neo for schools	ministry, city, school	NeoOS, Neo School, Classroom, Library and the sound library with no software fee; yearly support and updates, teacher training (2 days in person and a course in Neo School), a school channel in Renewal for software and settings	software €0; support €8 to 15 per device per year; training by agreement
Neo adaptation	a country or a language community	the right to a minimal adaptation together with the Neo team: a country pack (translation, voice, keyboards, holidays, knowledge libraries and the sound library in that language), the national layer in Renewal, a national Resident if wanted, the right to the "Neo" name for that version, and joint testing in pilot schools	one-off €150,000 to 300,000 (3 to 6 months, 4 to 6 Neo team members alongside a local team); maintenance €50,000 to 100,000 a year
Neo under licence	a local manufacturer	factory tools, the Neo test kit, the certificate, the right to the "Neo" name and a signed Neo ROM	a per-unit fee, by agreement (part of "Road to a standard")

What every offer guarantees, and what it does not offer

- All the code is public and reproducibly built; a country can check every byte of the image in ROM.
- The same privacy in every country: no hidden tracking, no keys to users' data, no exceptions for any authority. The national layer in Renewal carries software, certificates and translation; it cannot read or send users' files.
- News stays the user's choice: a country pack may *offer* local sources, but none is switched on in advance.

How an adaptation runs

1. **Enquiry and assessment** (2 weeks): language, script, schools, number of devices, and what already exists (translations in Weblate, voice models, Kiwix in that language).
2. **Contract** with a clear scope and deadlines.
3. **Joint team:** local translators and teachers work in a partner workspace (Weblate, with text reviewed in the context of the screen), and the Neo team reviews and merges.
4. **Pilot** in 5 to 10 schools, measured as in section 1 (10 minutes without a manual) in that language.
5. **Signing and delivery** of the country pack (below), then maintenance with every NeoOS release.

Team support as part of the OS

Neo Support is a system app: bug reports with logs only with consent (the user sees exactly what is sent), the status of a report, messages from the team, and expert remote help through "Call your own" (view only until the user says "yes", switches off after 30 minutes). For adaptation partners it also shows translation progress per part of the system and open questions. It works offline too: a report waits on the Worktable and is sent when a connection appears or through the Neighbourhood.

Protected delivery of localisation files

Step	How
Contents	A country pack is a neopak containing data only (strings, voice models, fonts, keyboards, libraries, music). Executable code is not allowed in it; code goes only through Treasury, with reproducible builds.
Two signatures	The Neo team and the partner each sign every pack with their keys in the Safe chip; without both signatures the pack will not install.
Encryption	before public release the pack is encrypted for its recipients (a school, a ministry, pilot devices), so an unfinished translation does not leak.
Three routes, one check	a **cartridge** for schools without internet, the **LAN** through the school's Treasury mirror in the Neighbourhood, or **any network connection** through the Renewal channel. On every route: signature and version check, who approved it, and an installation log.
Rollback	a new pack goes into the other half of the A/B image; if something is wrong, one click restores the previous version.

```
Software work: about 45 P-M (Neo Support 15, protected pack delivery 20,
partner tools 10), counted under Applications and Tools and ecosystem.
Prices are indicative, for discussion, not a formal quote.
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19. After 1.0

Version	Roughly	Brings
1.1	3 to 4 months after 1.0	Android room, Neo Workshop, Neo 100 and Neo OS Lite, P1 apps that did not make it, Neo Maps, Claude as an option for Ask Neo, the first fixes from real use
1.5	about 9 months after 1.0	Neo Studio, Cut, Broadcast, Stream across several Neo computers at home, Neo Revive for old x86 and ARM computers
2.0	about 18 months after 1.0	ReactOS box, Neo OS Lite for phones, export from Neo Studio to Android and iOS, a new generation of hardware, Ras if needed

After 1.0 the teams shrink to about 65 people for maintenance, new versions and developer support, with ten years of updates for every model, as promised in the Neo principles.

20. Glossary: English and Serbian names

NeoOS was designed in Serbian, and every name has an English counterpart chosen for meaning rather than sound. The user always sees the names of the chosen language; scripts use the technical names, which are the same in every language. Names marked UK/US differ between British and American English.

Area	English	Serbian
Kernel and core	Fortress, Gate, Dispatcher, Swarm, Chronos, Exec-N	Tvrđava, Kapija, Dispečer, Roj, Hronos, Exec-N
Services	Renewal, Lock, Safe, Keeper, Find, Datatypes, Undo, Chameleon	Obnova, Brava, Sef, Čuvar, Traži, Tipovi podataka, Poništi, Kameleon
Chips	Courier, Copernicus, Newton, Ray, Panorama, Orpheus, Distributor, Neuron	Kurir, Kopernik, Njutn, Zrak, Panorama, Orfej, Razvodnik, Neuron
Compatibility	Bridge, Link, Boxes, Satellite, Arcade, Treasury, Revive	Most, Spona, Boxovi, Satelit, Arkada, Riznica, Oživi
Applications	Cabinet, Horizon, Pigeon, Quartet (Quill, Abacus, Canvas, Stage), Archive, Leaf, Gramophone, Cinema, Wave, Diary, Talk, Keyring, Interpreter, Tether, Shell, Cut, Broadcast, News	Ormar, Vidik, Golub, Četvorka (Pero, Abakus, Platno, Pozornica), Arhiva, Listaj, Gramofon, Bioskop, Talas, Rokovnik, Razgovor, Ključar, Tumač, Veza, Školjka, Montaža, Emisija, Vesti
Learning	Ask Neo, Library, Neo School, Classroom, Playroom, Neo Blocks, Lab, Sky, Recall	Pitaj Neo, Biblioteka, Neo Škola, Učionica, Igraonica, Neo Kocke, Laboratorija, Nebo, Pamtilica
Workshop pack	Workshop, Sculptor, Brush, Vector, Type, Darkroom, Studio, Director	Radionica, Vajar, Kist, Vektor, Slog, Tamna komora, Studio, Režiser
Logical disks	System:, Apps:, Home:, Pads:, Settings:, Boxes:, RAM:, RAD:	Sistem:, Aplikacije:, Dom:, Blokovi:, Podešavanja:, Boxovi:, RAM:, RAD:
Easy-mode names	Desk, Letterbox (UK) / Mailbox (US), Bin (UK) / Trash (US), Foundation, Toolshed, Guest rooms, Notepad, Worktable, Caretaker, Keys, Neighbourhood (UK) / Neighborhood (US), World, Household, Kids' room, Back to yesterday, Nap, Good night, Templates	Sto, Sanduče, Kanta, Temelj, Alatnica, Gostinjske sobe, Blok, Radni sto, Domar, Ključevi, Komšiluk, Svet, Ukućani, Dečja soba, Vrati na juče, Dremka, Laku noć, Šabloni
Desktop	Shelves, Helpers, Tiles, Bar, Neo button, Meter, Harbour, Neo DOS, Shell, Linux Shell, Translator, Crunch, Decrunch, Disk tile, Cabinet Pro, Duo, button bank, Triggers, Mine menu, Tasks, Jump, Pocket, Case, Bag, Board, Co-op, Nook	Police, Pomagači, Pločice, Traka, Neo dugme, Merač, Pristanište, Neo DOS, Školjka, Linux Školjka, Prevodilac, Zgusni, Razgusni, pločica Disk, Ormar Pro, Dvojac, banka dugmića, Okidači, meni Moje, Zadaci, Skok, Džepić, Predmet, Torba, Tabla, Zadruga, Kutak
Styles and themes	Retro, Future Shock, Neo Silver, Slate, Calm, Lively, Lavish, Paper, Pixel Block, Glass, Classic, First Desk	Retro, Šok budućnosti, Neo Srebro, Škriljac, Mirno, Živo, Raskošno, Papir, Kockica, Staklo, Klasik, Prvi Sto
Device profiles	Pocket, Light, Handheld Pro, Home, Workstation, Large, Shared computer, Full throttle	Džepni, Lagani, Ručni jaki, Kućni, Radna stanica, Uvećano, Zajednički računar, Pun gas
Games	Crystal, Grandmaster, Midnight Poker, Pinball Cosmos, Forest Bane, Swarms, Depths	Kristal, Velemajstor, Ponoćni poker, Fliper Kosmos, Šumski Bane, Rojevi, Dubine
Sound and support	Neo Sound, Build it yourself, sound library, Read to me, family library, country pack, Neo Support	Neo Zvuk, Sklopi sam, Fonoteka, Čitaj mi, Porodična biblioteka, Paket zemlje, Neo Podrška
Characters	Dijana, Ras, Residents	Dijana, Ras, Stanari